

(12)
United States Patent
Fu et al.

(10) **Patent No.:** **US 12,400,661 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54)
SYSTEMS AND METHODS FOR CAPTURING, PROCESSING, AND RENDERING ONE OR MORE CONTEXT-AWARE MOMENT-ASSOCIATING ELEMENTS

(71)
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(*)
Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21)
Appl. No.: **18/514,666**

(22)
Filed: **Nov. 20, 2023**

(65)
Prior Publication Data
US 2024/0087574 A1 Mar. 14, 2024

Related U.S. Application Data
(63)
Continuation of application No. 17/242,465, filed on Apr. 28, 2021, now Pat. No. 11,869,508, which is a (Continued)

(51)
Int. Cl.
G10L 15/26 (2006.01)
G06F 16/906 (2019.01)
(Continued)

(52)
U.S. Cl.
CPC **G10L 15/26** (2013.01); **G06F 16/906** (2019.01); **G10L 15/04** (2013.01); **G10L 17/00** (2013.01)

(58)
Field of Classification Search
CPC G10L 15/26; G10L 15/04; G10L 17/00; G10L 15/32; G10L 15/00; G06F 16/906; (Continued)

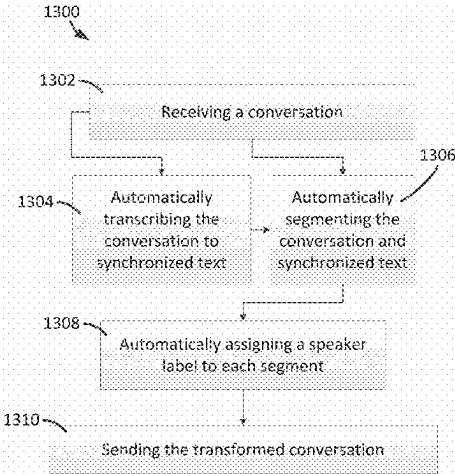
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(57)
ABSTRACT
Computer-implemented method and system for receiving and processing one or more moment-associating elements. For example, the computer-implemented method includes receiving the one or more moment-associating elements, transforming the one or more moment-associating elements into one or more pieces of moment-associating information, and transmitting at least one piece of the one or more pieces of moment-associating information. The transforming the one or more moment-associating elements into one or more (Continued)



pieces of moment-associating information includes segmenting the one or more moment-associating elements into a plurality of moment-associating segments, assigning a segment speaker for each segment of the plurality of moment-associating segments, transcribing the plurality of moment-associating segments into a plurality of transcribed segments, and generating the one or more pieces of moment-associating information based on at least the plurality of transcribed segments and the segment speaker assigned for each segment of the plurality of moment-associating segments.

20 Claims, 15 Drawing Sheets

Related U.S. Application Data

continuation of application No. 16/403,263, filed on May 3, 2019, now Pat. No. 11,024,316, and a continuation-in-part of application No. 16/027,511, filed on Jul. 5, 2018, now Pat. No. 10,978,073, said application No. 16/403,263 is a continuation-in-part of application No. 16/276,446, filed on Feb. 14, 2019, now Pat. No. 11,100,943, which is a continuation-in-part of application No. 16/027,511, filed on Jul. 5, 2018, now Pat. No. 10,978,073.

- (60) Provisional application No. 62/668,623, filed on May 8, 2018, provisional application No. 62/631,680, filed on Feb. 17, 2018, provisional application No. 62/710,631, filed on Feb. 16, 2018, provisional application No. 62/530,227, filed on Jul. 9, 2017.

(51) **Int. Cl.**

G10L 15/04 (2013.01)

G10L 17/00 (2013.01)

(58) **Field of Classification Search**

CPC G06F 16/685; G06F 16/683; H04M 3/567; H04M 3/56; H04M 3/42127; H04M 3/42008; H04M 3/58; H04M 3/42391; H04M 3/42; H04M 2201/40; H04M 2201/22; H04M 2201/38; H04L 12/1822; H04L 12/18; H04L 51/066; H04L 12/1831

See application file for complete search history.

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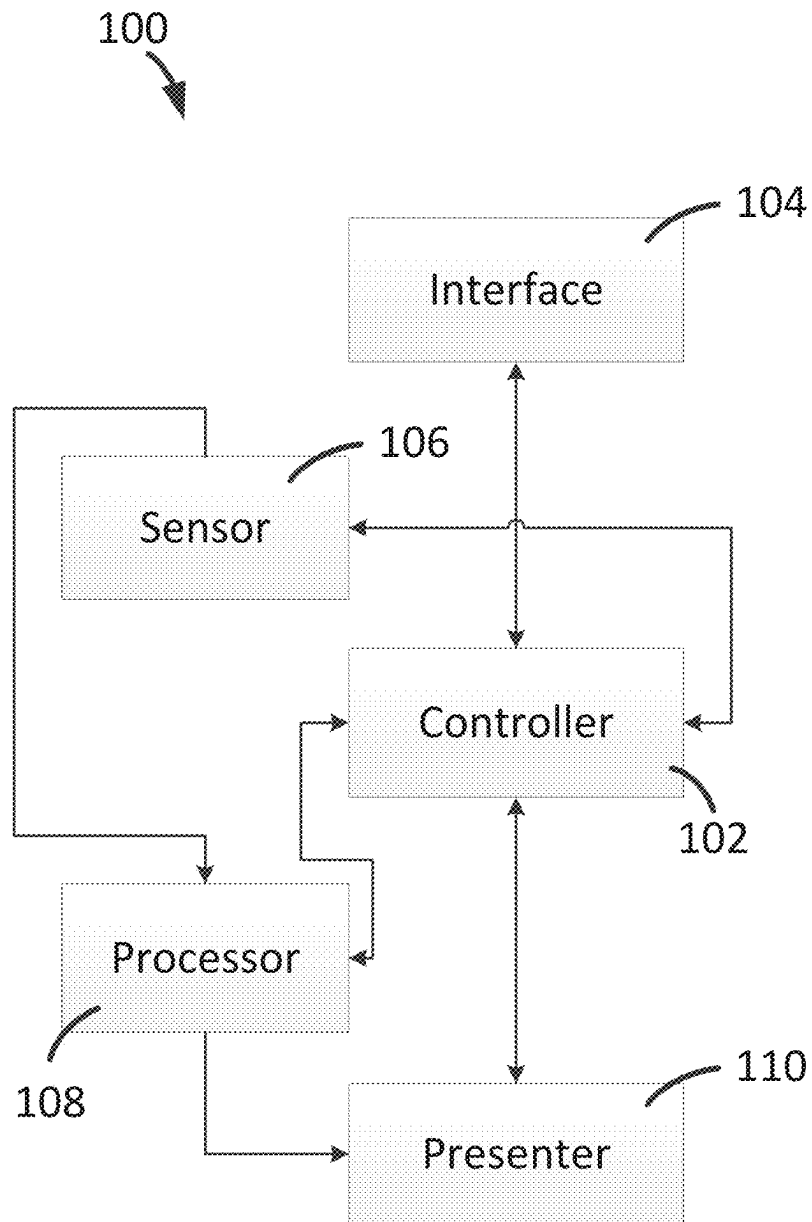


FIG. 1

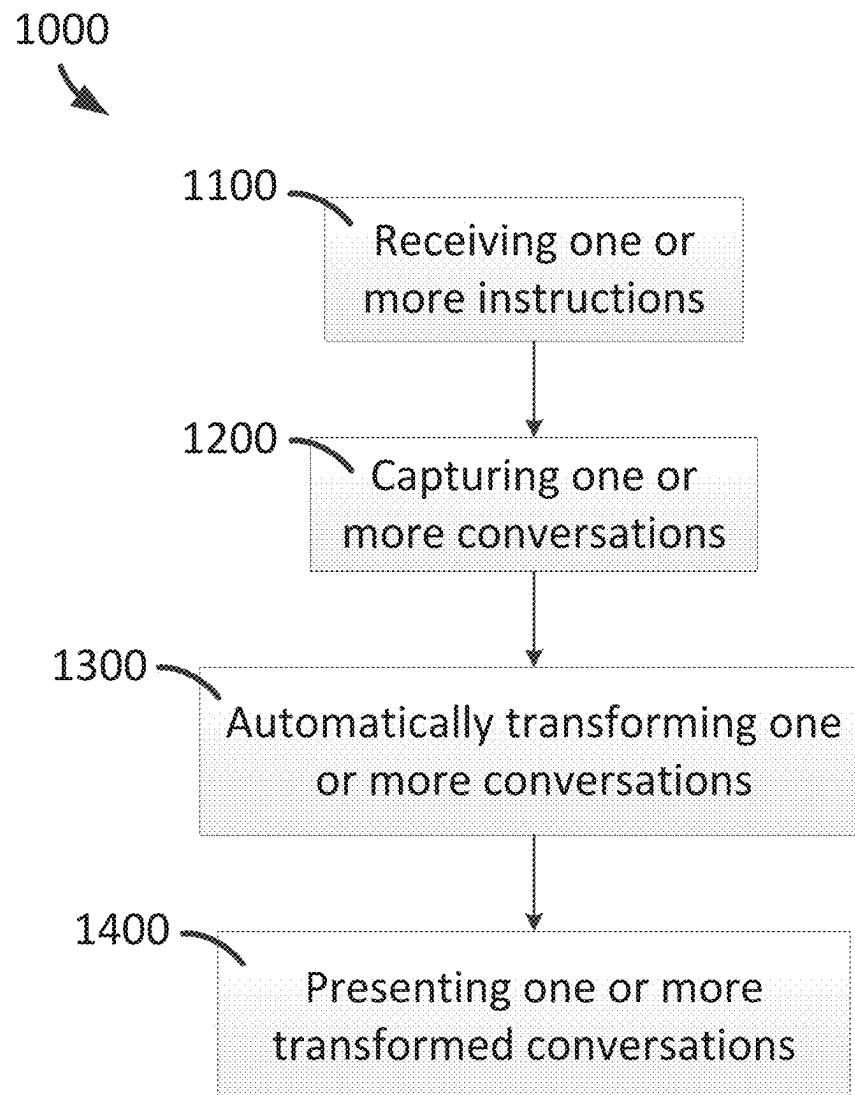


FIG. 2

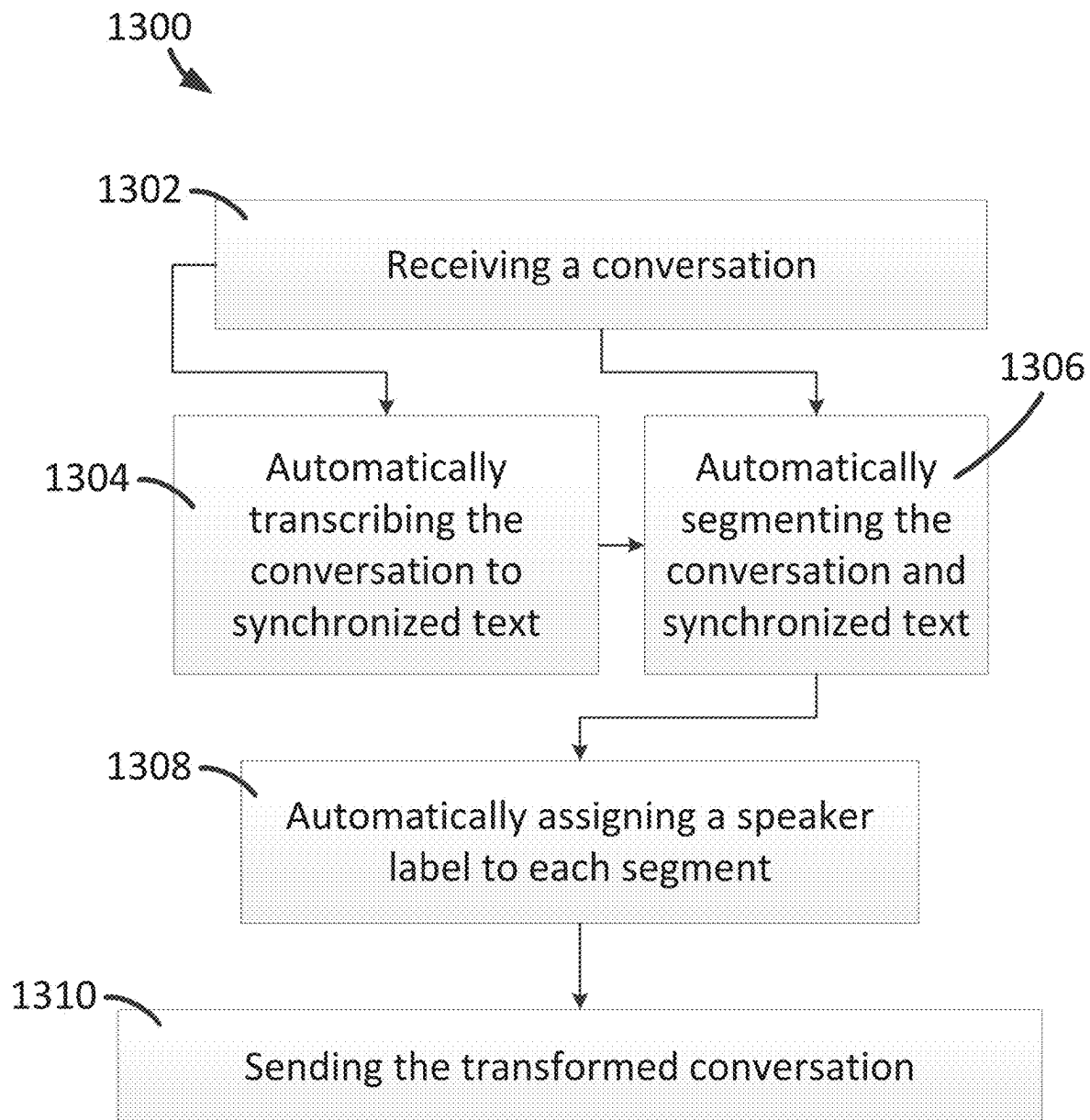


FIG. 3

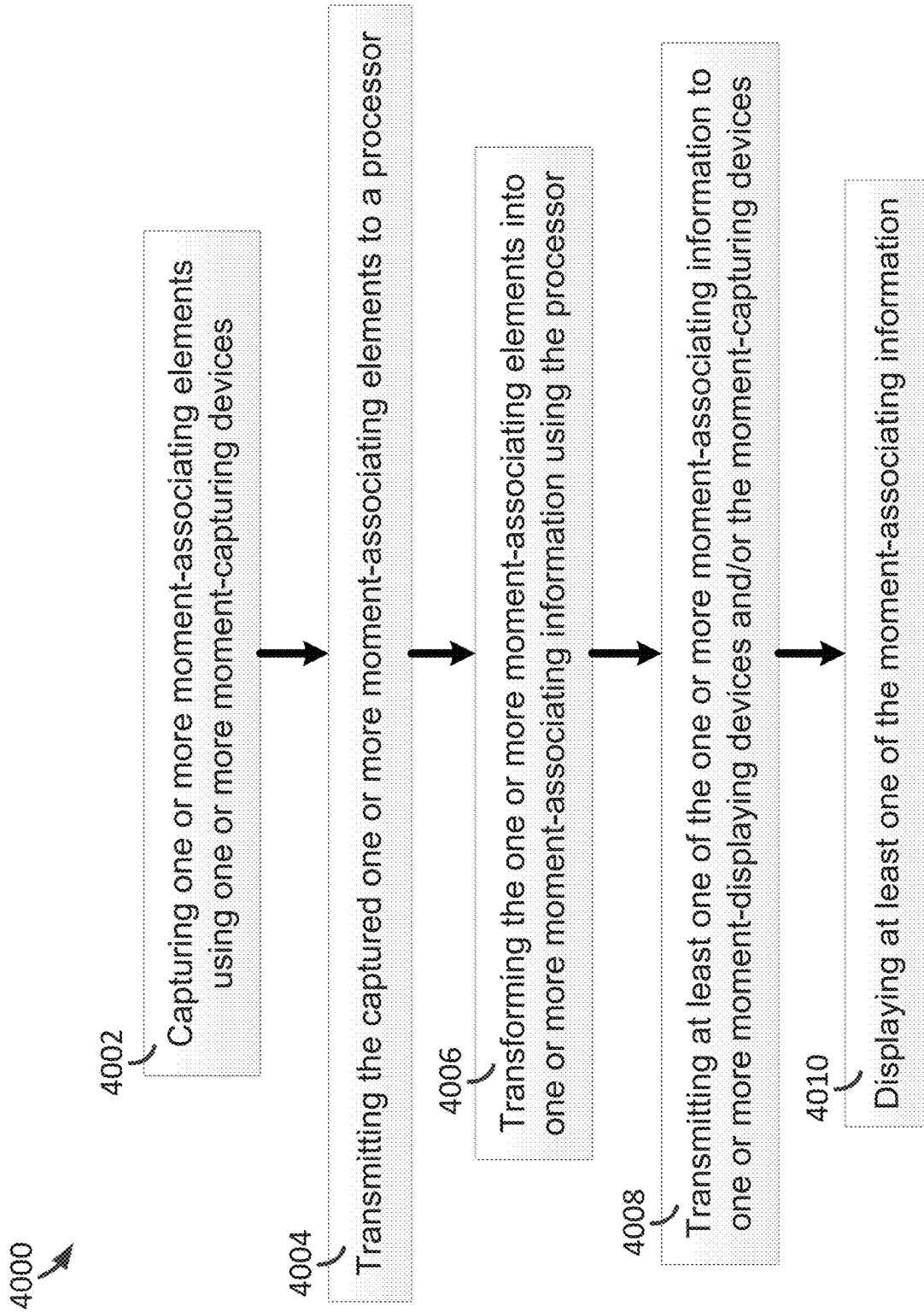


FIG. 4

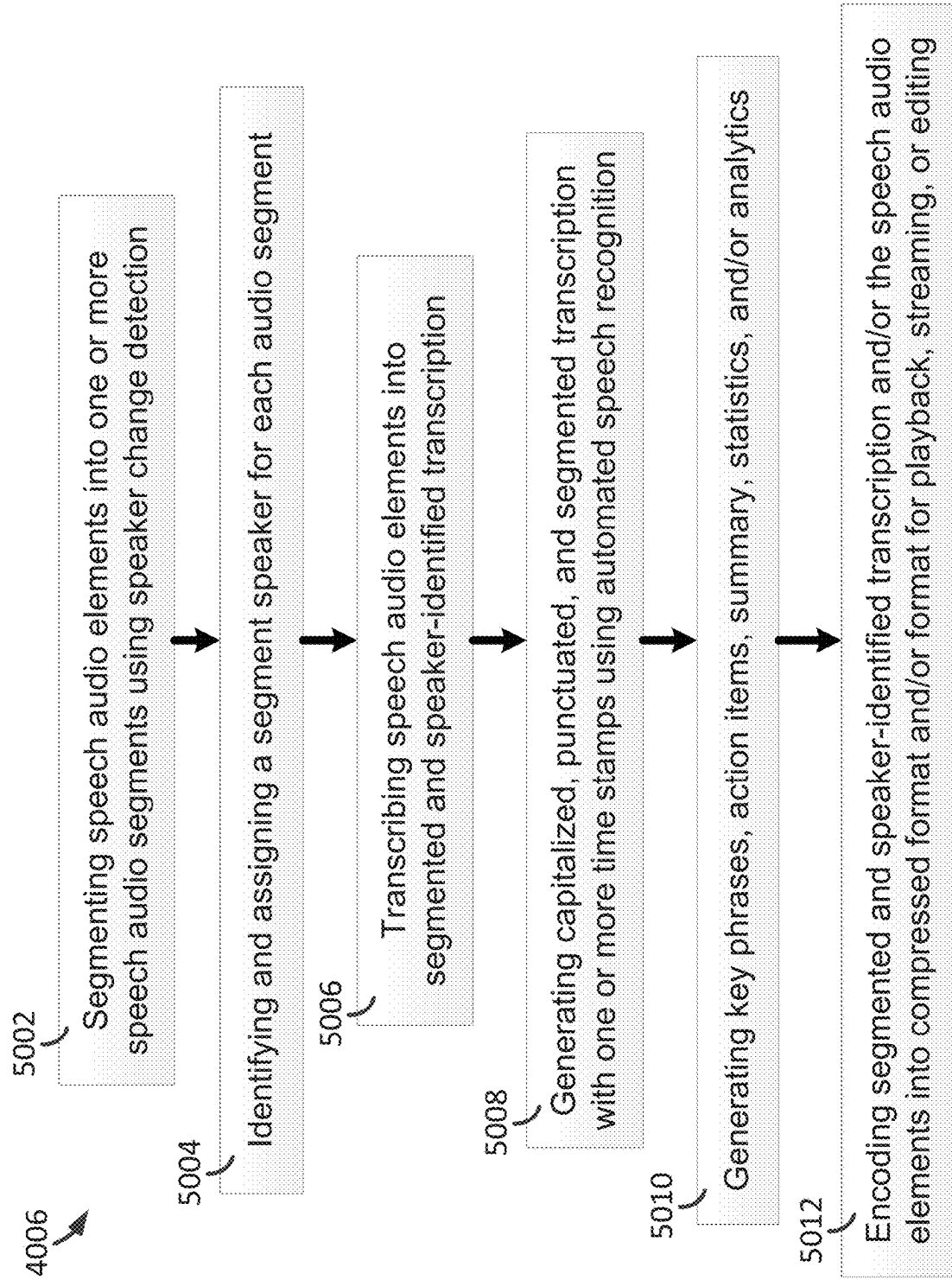


FIG. 5

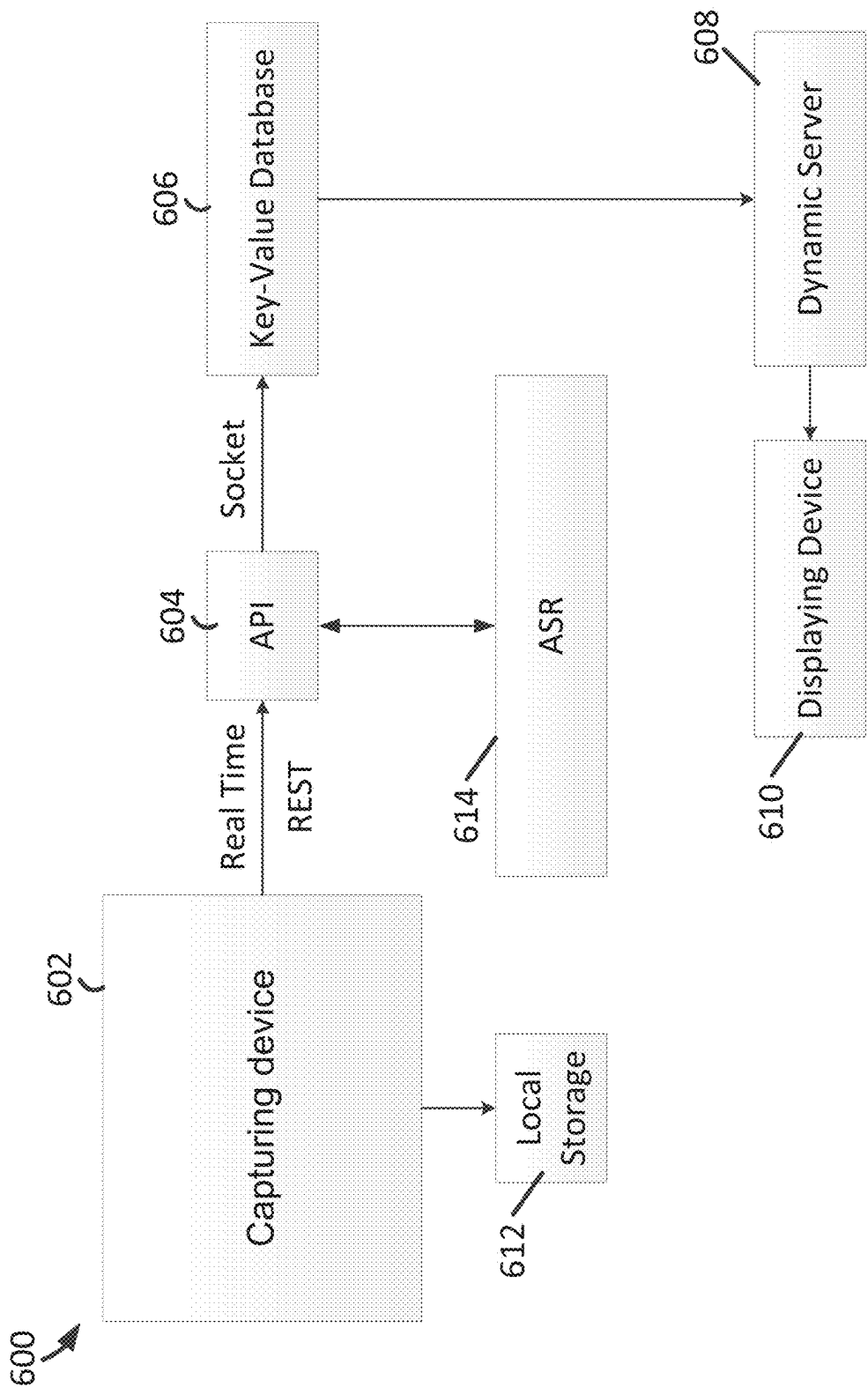


FIG. 6

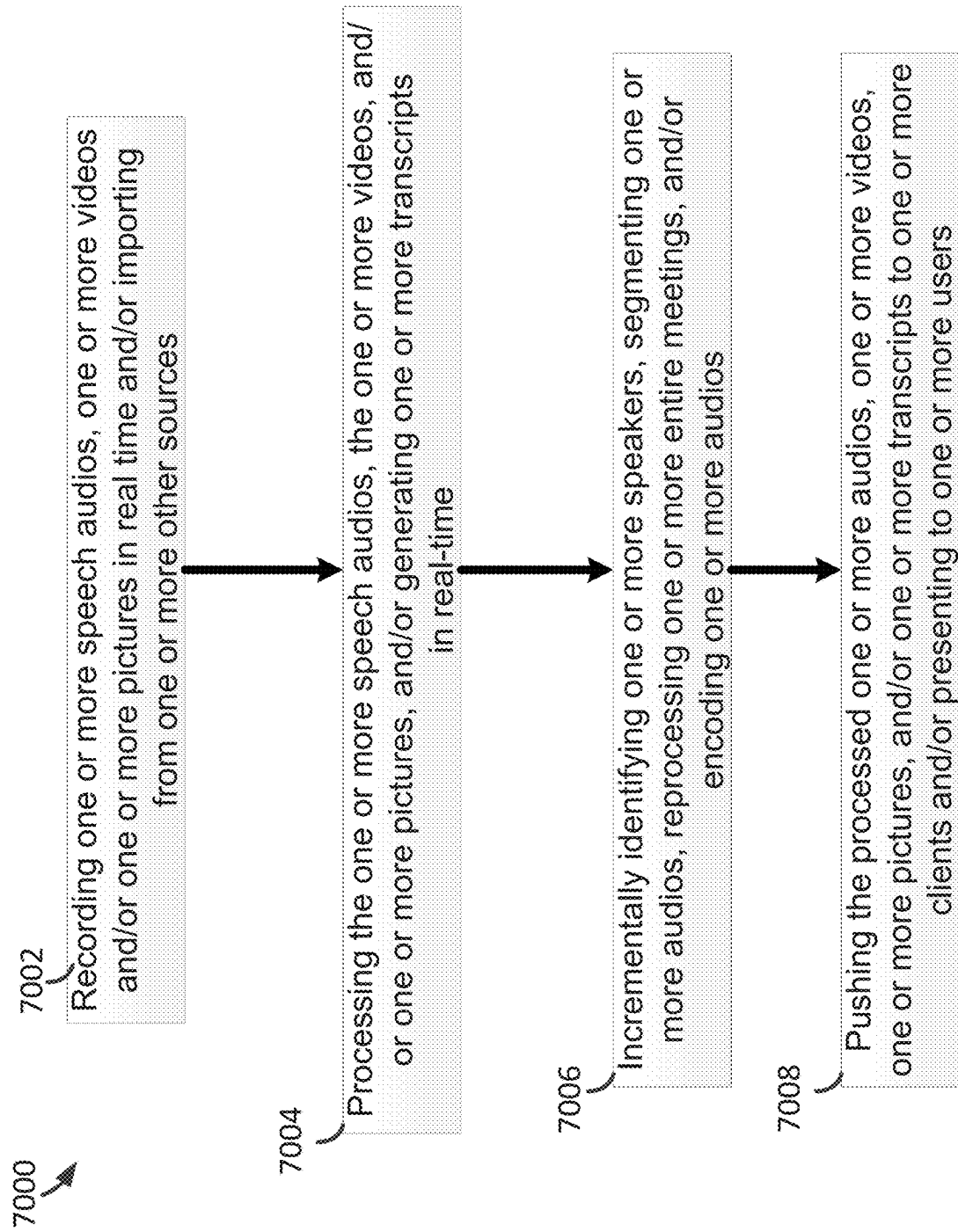


FIG. 7

8000 ↗

8002

Capturing one or more speeches and saving to one or more local disks



8004

Establishing one or more persistent connections with one or more servers to upload one or more audios and/or getting one or more transcript updates in real time

FIG. 8

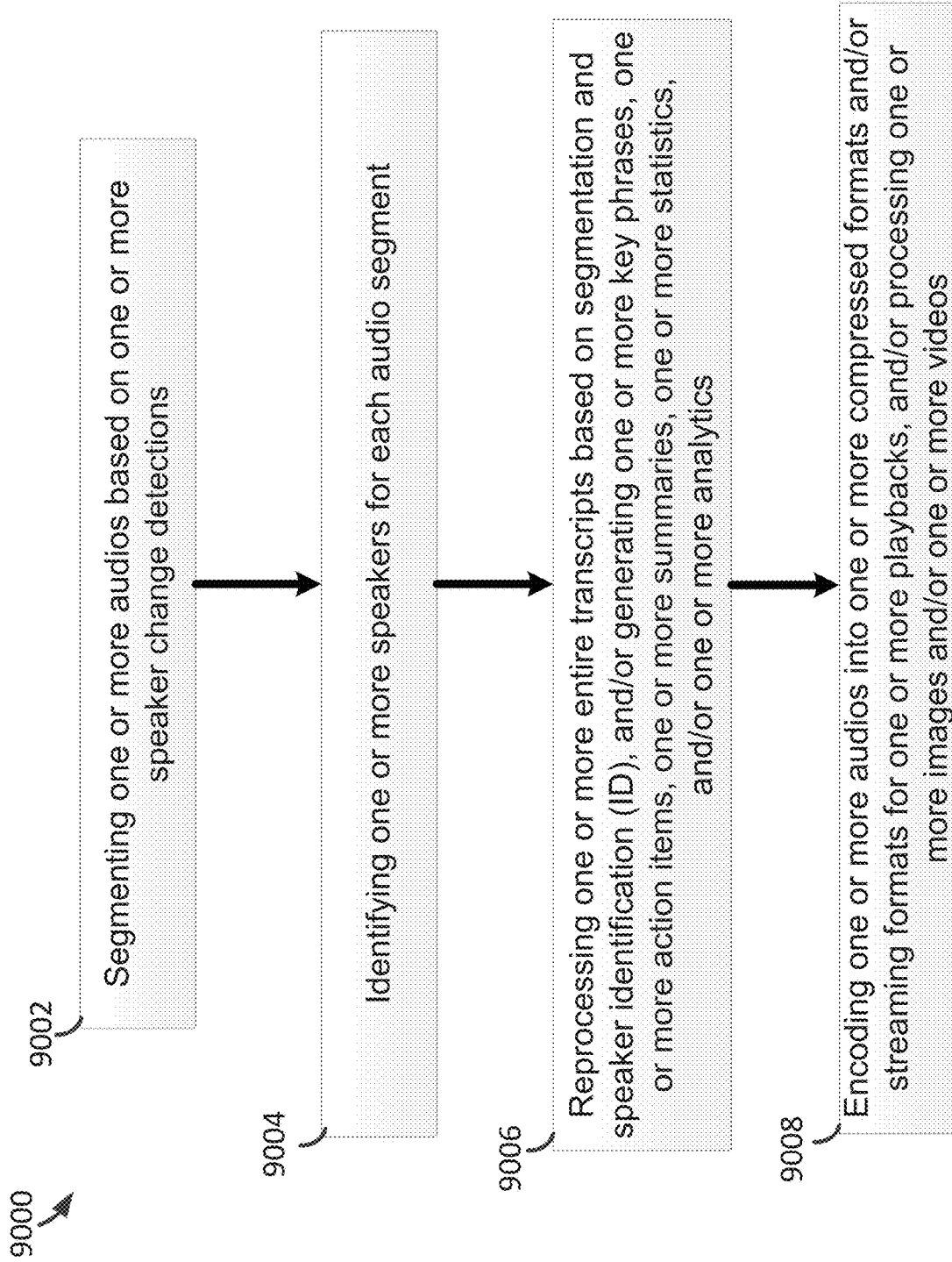


FIG. 9

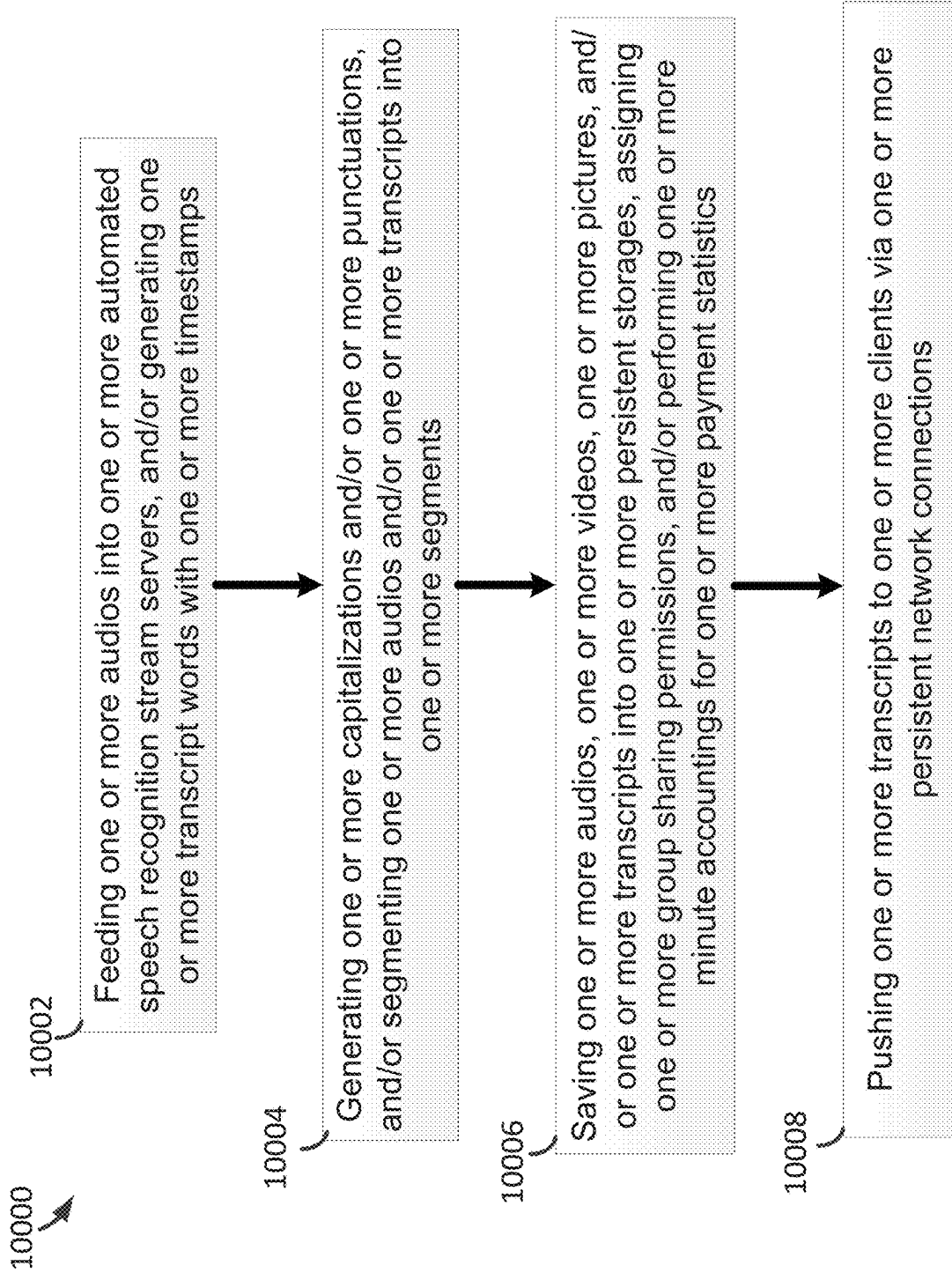


FIG. 10

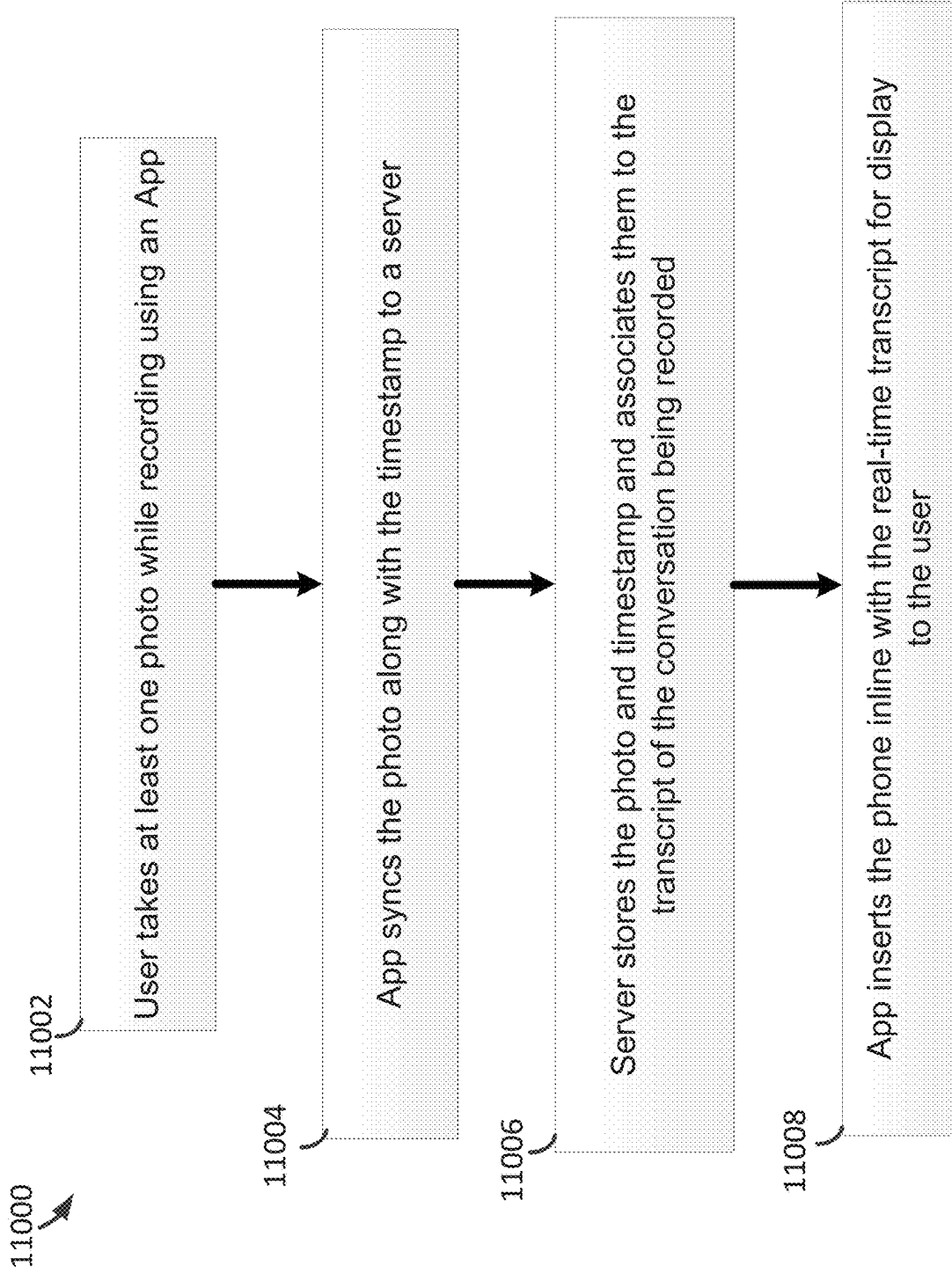


FIG. 11

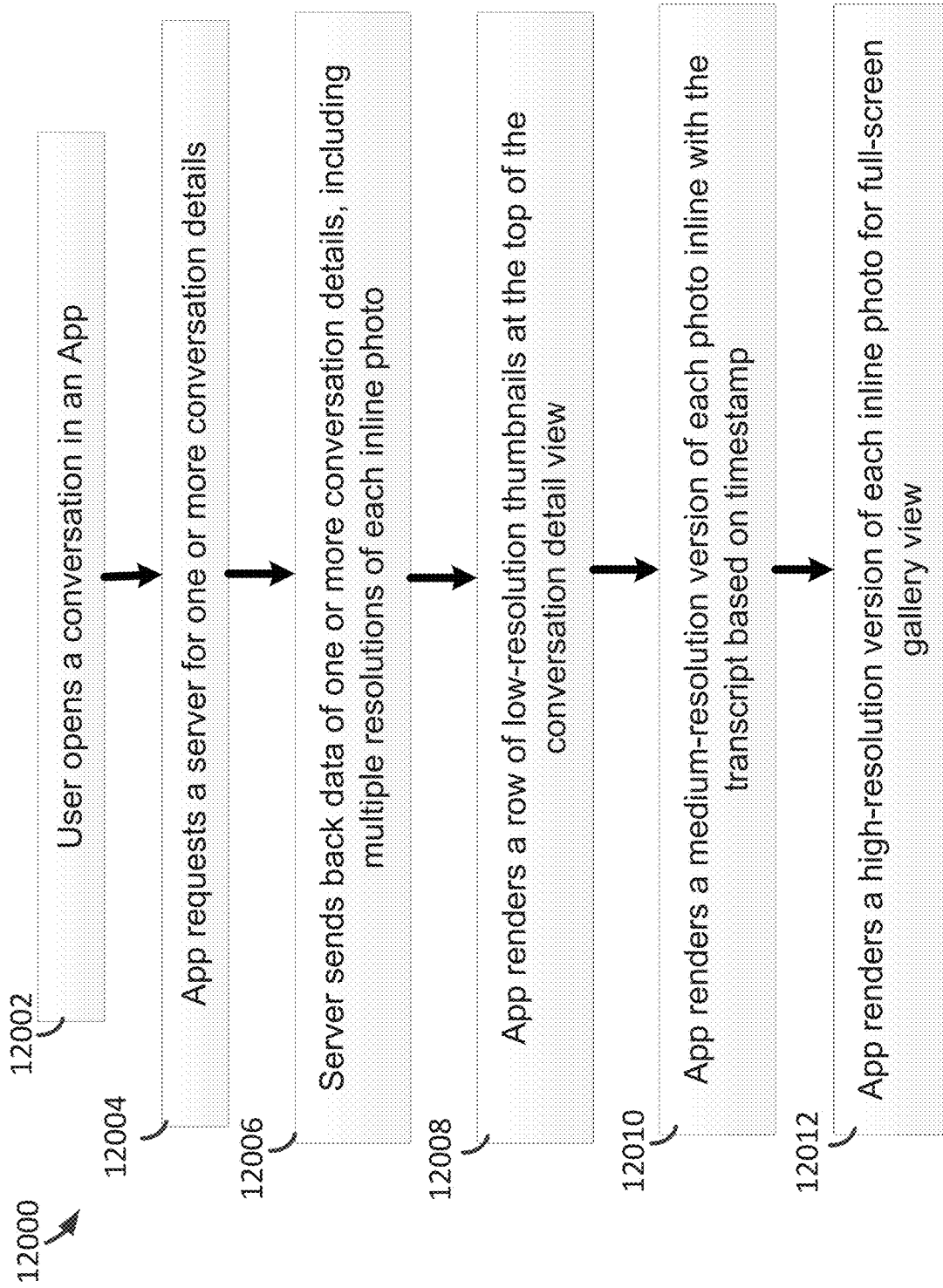


FIG. 12

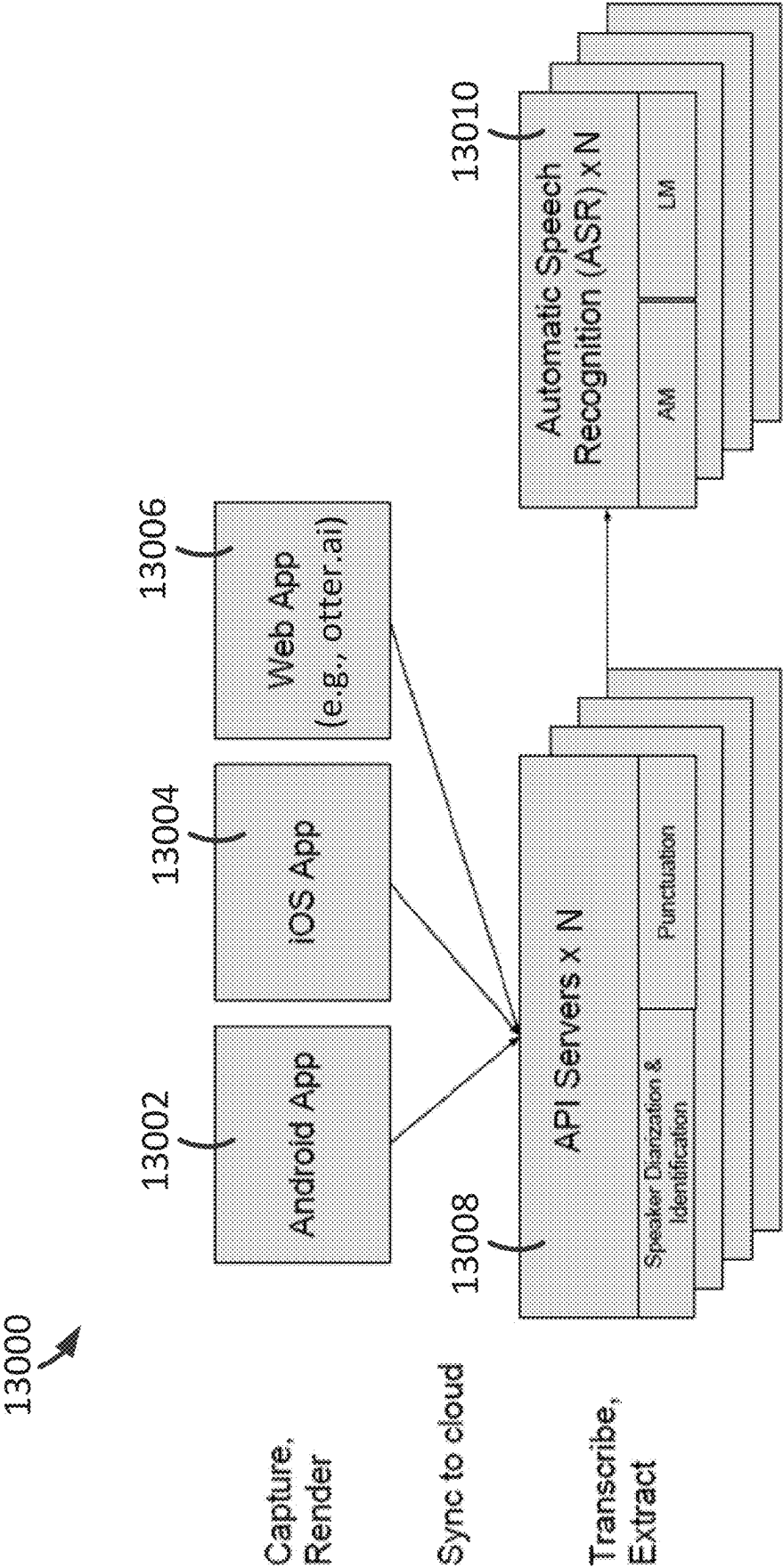


FIG. 13

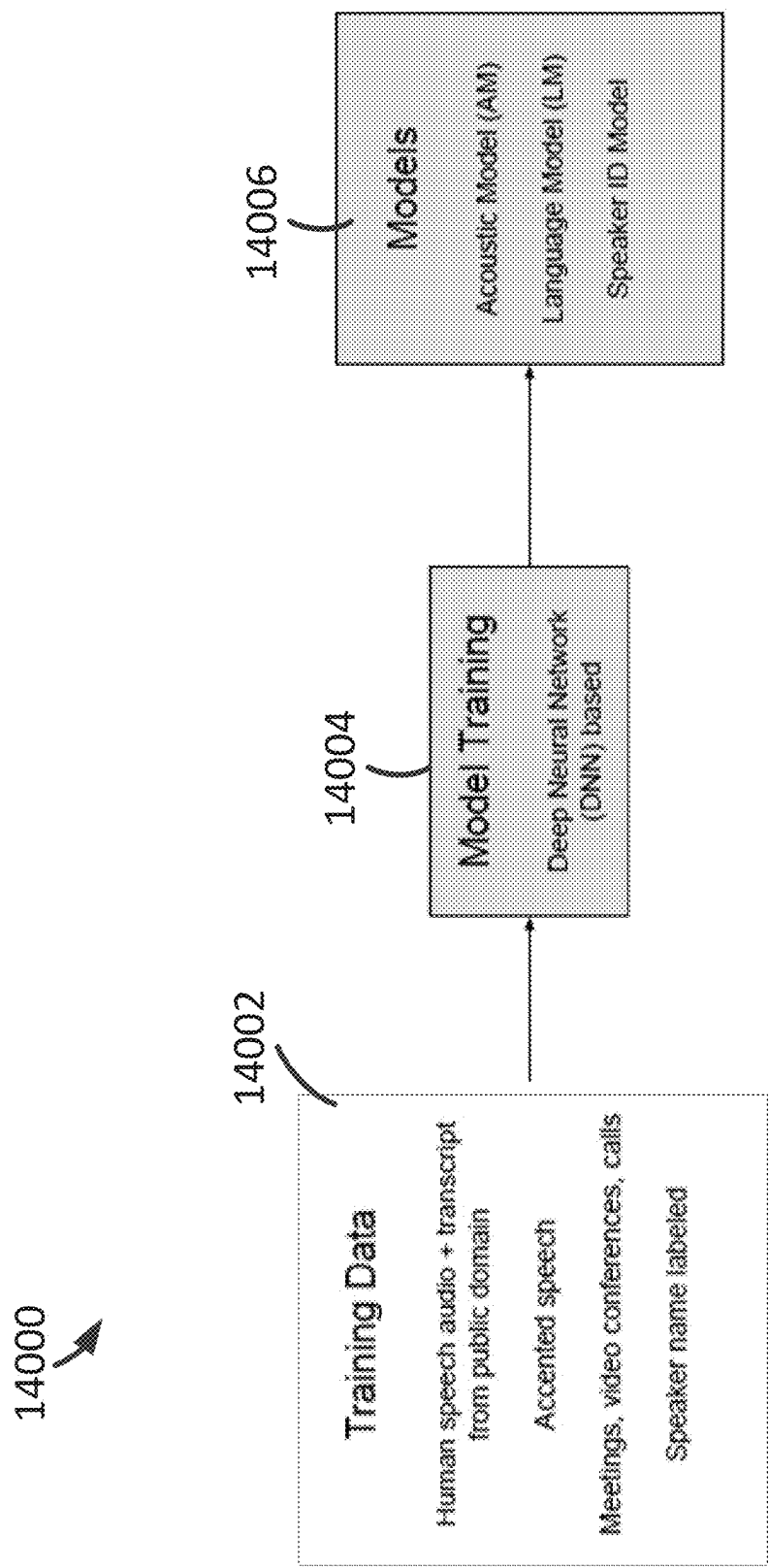


FIG. 14

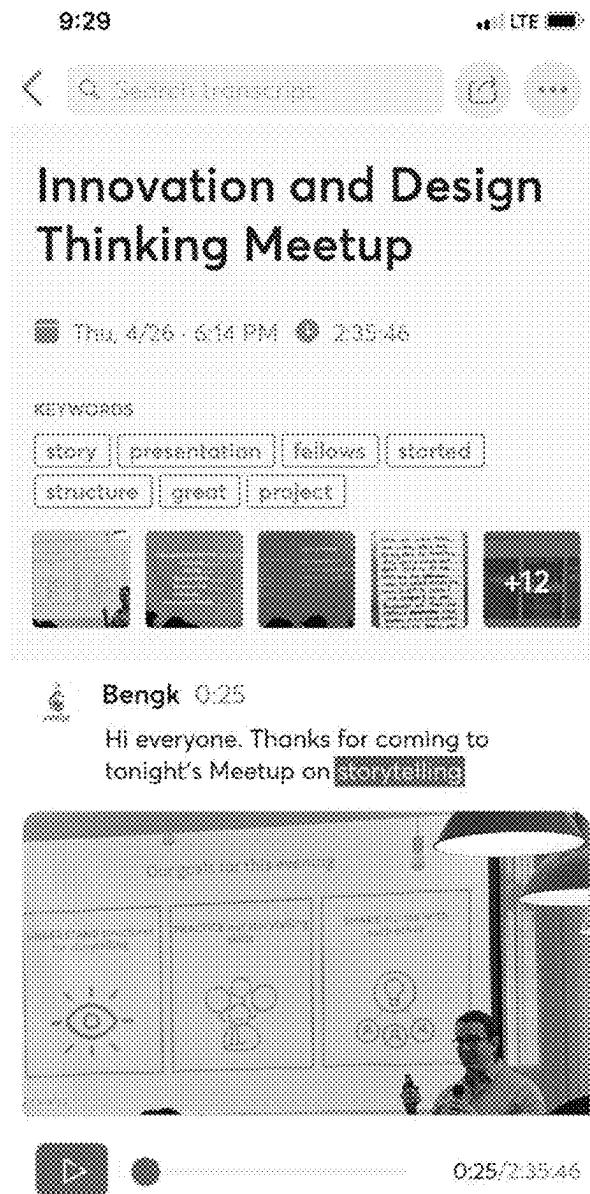


FIG. 15

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SYSTEMS AND METHODS FOR CAPTURING, PROCESSING, AND RENDERING ONE OR MORE CONTEXT-AWARE MOMENT-ASSOCIATING ELEMENTS

CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/242,465, filed Apr. 28, 2021, which is a continuation of U.S. patent application Ser. No. 16/403,263, filed May 3, 2019, which claims priority to U.S. Provisional Patent Application No. 62/668,623, filed May 8, 2018, both of these applications being incorporated by reference herein for all purposes. In addition, U.S. patent application Ser. No. 16/403,263 is a continuation-in-part of U.S. patent application Ser. No. 16/027,511, filed Jul. 5, 2018, claiming priority to U.S. Provisional Patent Application No. 62/530,227, filed Jul. 9, 2017, all of these applications being incorporated by reference herein for all purposes. Moreover, U.S. patent application Ser. No. 16/403,263 is a continuation-in-part of U.S. patent application Ser. No. 16/276,446, filed Feb. 14, 2019, which claims priority to U.S. Provisional Patent Application No. 62/710,631, filed Feb. 16, 2018, U.S. Provisional Patent Application No. 62/631,680, filed Feb. 17, 2018, and U.S. Provisional Patent Application No. 62/668,623, filed May 8, 2018, and is a continuation-in-part of U.S. patent application Ser. No. 16/027,511, filed Jul. 5, 2018, claiming priority to U.S. Provisional Patent Application No. 62/530,227, filed Jul. 9, 2017, all of these applications being incorporated by reference herein for all purposes.

BACKGROUND OF THE INVENTION

Certain embodiments of the present invention are directed to signal processing. More particularly, some embodiments of the invention provide systems and methods for capturing, processing, and rendering one or more context-aware moment-associating elements such as one or more speeches and/or one or more photos. Merely by way of example, some embodiments of the invention have been applied to conversations captured in audio form. But it would be recognized that the invention has a much broader range of applicability.

Conversations, such as human-to-human conversations, include information that is often difficult to comprehensively, efficiently, and accurately extract, using conventional methods and systems. For example, conventional note-taking performed during a conversation not only distracts the note-taker from the conversation but can also lead to inaccurate recordation of information due to human-error, such as for human's inability to multitask well and process information efficiently with high accuracy in real time.

Hence it is highly desirable to provide systems and methods for capturing, processing, and rendering conversations (e.g., in an automatic manner) to increase the value of conversations, such as human-to-human conversations, at least by increasing the comprehensiveness and accuracy of information extractable from the conversations.

BRIEF SUMMARY OF THE INVENTION

Certain embodiments of the present invention are directed to signal processing. More particularly, some embodiments of the invention provide systems and methods for capturing, processing, and rendering one or more context-aware moment-associating elements such as one or more speeches

2

and/or one or more photos. Merely by way of example, some embodiments of the invention have been applied to conversations captured in audio form. But it would be recognized that the invention has a much broader range of applicability.

According to one embodiment, a computer-implemented method for receiving and processing one or more moment-associating elements includes receiving the one or more moment-associating elements, transforming the one or more moment-associating elements into one or more pieces of moment-associating information, and transmitting at least one piece of the one or more pieces of moment-associating information. The transforming the one or more moment-associating elements into one or more pieces of moment-associating information includes segmenting the one or more moment-associating elements into a plurality of moment-associating segments, assigning a segment speaker for each segment of the plurality of moment-associating segments, transcribing the plurality of moment-associating segments into a plurality of transcribed segments, and generating the one or more pieces of moment-associating information based on at least the plurality of transcribed segments and the segment speaker assigned for each segment of the plurality of moment-associating segments.

According to another embodiment, a system for receiving and processing one or more moment-associating elements includes a receiving module configured to receive the one or more moment-associating elements, a transforming module configured to transform the one or more moment-associating elements into one or more pieces of moment-associating information, and a transmitting module configured to transmit at least one piece of the one or more pieces of moment-associating information. The transforming module is further configured to segment the one or more moment-associating elements into a plurality of moment-associating segments, assign a segment speaker for each segment of the plurality of moment-associating segments, transcribe the plurality of moment-associating segments into a plurality of transcribed segments, and generate the one or more pieces of moment-associating information based on at least the plurality of transcribed segments and the segment speaker assigned for each segment of the plurality of moment-associating segments.

According to yet another embodiment, a non-transitory computer-readable medium with instructions stored thereon, that when executed by a processor, perform the processes comprising: receiving the one or more moment-associating elements; transforming the one or more moment-associating elements into one or more pieces of moment-associating information; and transmitting at least one piece of the one or more pieces of moment-associating information. The transforming the one or more moment-associating elements into one or more pieces of moment-associating information includes: segmenting the one or more moment-associating elements into a plurality of moment-associating segments; assigning a segment speaker for each segment of the plurality of moment-associating segments; transcribing the plurality of moment-associating segments into a plurality of transcribed segments; and generating the one or more pieces of moment-associating information based on at least the plurality of transcribed segments and the segment speaker assigned for each segment of the plurality of moment-associating segments.

Depending upon embodiment, one or more benefits may be achieved. These benefits and various additional objects, features and advantages of the present invention can be fully appreciated with reference to the detailed description and accompanying drawings that follow.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a simplified diagram showing a system for processing and presenting one or more conversations according to some embodiments of the present invention.

FIG. 2 is a simplified diagram showing a method for processing and presenting one or more conversations according to some embodiments of the present invention.

FIG. 3 is a simplified diagram showing the process for automatically transforming one or more conversations as shown in FIG. 2 according to some embodiments of the present invention.

FIG. 4 is a simplified diagram showing a method of capturing and displaying one or more moments according to certain embodiments of the present invention.

FIG. 5 is a simplified diagram showing process for transforming the one or more moment-associating elements into one or more moment-associating information as shown in FIG. 4 according to some embodiments of the present invention.

FIG. 6 is a simplified diagram showing a system for capturing, processing, and rendering a context-aware moment-associating element according to certain embodiments of the present invention.

FIG. 7 is a simplified diagram showing a method of operation for the system as shown in FIG. 6 according to some embodiments of the present invention.

FIG. 8 is a simplified diagram showing a process of operation for front-end and/or client related to FIG. 7 according to certain embodiments of the present invention.

FIG. 9 is a simplified diagram showing a process of operation for backend incremental processing related to FIG. 7 according to some embodiments of the present invention.

FIG. 10 is a simplified diagram showing a process of operation for backend real-time processing related to FIG. 7 according to certain embodiments of the present invention.

FIG. 11 is a simplified diagram showing a method for real-time capturing of one or more inline photos according to some embodiments of the present invention.

FIG. 12 is a simplified diagram showing a method for rendering of one or more inline photos according to certain embodiments of the present invention.

FIG. 13 is a simplified diagram showing a system for real-time capture, processing, and rendering of one or more context-aware conversations according to some embodiments of the present invention.

FIG. 14 is a simplified diagram showing a method of training the system as shown in FIG. 13 according to certain embodiments of the present invention.

FIG. 15 is a simplified diagram showing an example of user interface for one or more inline photos related to the method as shown in FIG. 11 and/or the method as shown in FIG. 12 according to some embodiments of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

Certain embodiments of the present invention are directed to signal processing. More particularly, some embodiments of the invention provide systems and methods for capturing, processing, and rendering one or more context-aware moment-associating elements such as one or more speeches and/or one or more photos. Merely by way of example, some embodiments of the invention have been applied to conver-

sations captured in audio form. But it would be recognized that the invention has a much broader range of applicability.

FIG. 1 is a simplified diagram showing a system for processing and presenting one or more conversations according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The system 100 includes a controller 102, an interface 104, a sensor 106, a processor 108, and a presenter 110. In some examples, the presenter 110 includes a mobile device, a web browser, a computer, a watch, a phone, a tablet, a robot, a projector, a television, and/or a display. In certain examples, the presenter 110 includes part of a mobile device, part of a web browser, part of a computer, part of a watch, part of a phone, part of a tablet, part of a robot, part of a projector, part of a television, and/or part of a display. Although the above has been shown using a selected group of components for the system, there can be many alternatives, modifications, and variations. For example, some of the components may be expanded and/or combined. Other components may be inserted to those noted above. Depending upon the embodiment, the arrangement of components may be interchanged with others replaced. Further details of these components are found throughout the present specification.

In some embodiments, the controller 102 is configured to receive and/or send one or more instructions to other components of the system 100. For example, the controller 102 is configured to receive a first instruction from the interface 104 and send a second instruction to the sensor 106. In some examples, the controller 102 is or is part of a computing device (e.g., a computer, a phone, a laptop, a tablet, a watch, a television, a recording device, and/or a robot). In some embodiments, the controller includes hardware (e.g., a processor, a memory, a transmitter, and/or a receiver) and/or software for receiving, transmitting, and/or transforming instructions.

According to some embodiments, the interface 104 includes a user interface and/or is configured to receive a user instruction from a user of the system 100, and send a system instruction to one or more other components of the system 100 (e.g., the controller 102). For example, the interface includes a touchscreen, a button, a keyboard, a dialer (e.g., with number pad), an audio receiver, a gesture receiver, an application such as Otter for IOS or Android, and/or a webpage. In another example, the user is a human or another hardware and/or software system. In some embodiments, the interface 104 is configured to receive a first start instruction (e.g., when a user taps a start-record button in a mobile application) and to send a second start instruction to the controller 102 which in turn sends a third start instruction to, for example, the sensor 106. In some embodiments, the interface 104 is controlled by the controller 102 to provide one or more selectable actions (e.g., by the user). For example, the controller 102 controls the interface 104 to display a search bar and/or a record button for receiving instructions such as user instructions. In some embodiments, the interface 104 is communicatively coupled to the controller 102 and/or structurally contained or included in a common device (e.g., a phone).

In some embodiments, the sensor 106 is configured to receive an instruction and sense, receive, collect, detect, and/or capture a conversation in audio form (e.g., an audio file and/or an audio signal). For example, the sensor 106 includes an audio sensor and is configured to capture a conversation in audio form, such as to record a conversation

5

(e.g., a human-to-human conversation). In some examples, the audio sensor is a microphone, which is included as part of a device (e.g., a mobile phone) and/or a separate component coupled to the device (e.g., the mobile phone), and the device (e.g., the mobile phone) includes one or more components of the system 100 (e.g., controller 102). In some examples, the human-to-human conversation captured by the sensor 106 is sent (e.g., transmitted) to other components of the system 100. For example, the audio-form conversation captured by the sensor 106 (e.g., the audio recorded by the sensor 106) is sent to the processor 108 of the system 100. In some embodiments, the sensor 106 is communicatively coupled to the controller 102 such that the sensor 106 is configured to send a status signal (e.g., a feedback signal) to the controller 102 to indicate whether the sensor 106 is on (e.g., recording or capturing) or off (e.g., not recording or not capturing).

According to some embodiments, the processor 108 is configured to receive input including data, signal, and/or information from other components of the system 100, and to process, transform, transcribe, extract, and/or summarize the received input (e.g., audio recording). In some examples, the processor 108 is further configured to send, transmit, and/or present the processed output (e.g., transformed conversation). For example, the processor 108 is configured to receive the captured audio-form conversation (e.g., the audio recorded by the sensor 106) from the sensor 106. As an example, the processor 108 is configured to receive the conversation in audio form (e.g., an audio file and/or an audio signal) from the sensor 106. In some examples, the processor 108 is configured to be controlled by the controller 102, such as to process the data, signal, and/or information transmitted by the sensor 106, when an instruction sent from the controller 102 is received by the processor 108. In some embodiments, the processor 108 includes an automated speech recognition (ASR) system that is configured to automatically transform and/or transcribe a conversation (e.g., a captured conversation sent from the sensor 106), such as transforming the conversation from audio recording to synchronized transcription.

In some embodiments, the processor 108 is communicatively coupled to the controller 102 such that the processor 108 is configured to send a status signal (e.g., a feedback signal) to the controller 102 to indicate whether the processor 108 is processing or idling and/or to indicate a progress of a processing job. In some examples, the processor 108 includes an on-board processor of a client device such as a mobile phone, a tablet, a watch, a wearable, a computer, a television, and/or a robot. In some examples, the processor 108 includes an external processor of a server device and/or an external processor of another client device, such that the capturing (e.g., by the sensor 106) and the processing (e.g., by the processor 108) of the system 100 are performed with more than one device. For example, the sensor 106 is a microphone on a mobile phone (e.g., located at a client position) and is configured to capture a phone conversation in audio form, which is transmitted (e.g., wirelessly) to a server computer (e.g., located at a server position). For example, the server computer (e.g., located at a server position) includes the processor 108 configured to process the input (e.g., an audio file and/or an audio signal) that is sent by the sensor 106 and received by the processor 108.

According to some embodiments, the processor 108 is configured to output processed data, signal, and/or information, to the presenter 110 (e.g., a display) of the system 100. In some examples, the output is a processed or transformed form of the input received by the processor 108 (e.g., an

6

audio file and/or an audio signal sent by the sensor 106). For example, the processor 108 is configured to generate a transformed conversation and send the transformed conversation to the presenter 110 (e.g., a display) of the system 100. As an example, the processor 108 is configured to output synchronized text accompanied by a timestamped audio recording by transforming the conversation that is captured in audio form (e.g., captured by the sensor 106). In some embodiments, the processing and/or transforming performed by the processor 108 is real-time or near real-time. In some embodiments, the processor 108 is configured to process a live recording (e.g., a live recording of a human-to-human conversation) and/or a pre-recording (e.g., a pre-recording of a human-to-human conversation).

In some embodiments, the presenter 110 is configured to present, display, play, project, and/or recreate the conversation that is captured, for example, by the sensor 106, before and/or after transformation by the processor 108. For example, the presenter 110 (e.g., a display) is configured to receive the transformed conversation from the processor 108 and present the transformed conversation. As an example, the presenter 110 (e.g., a display) receives the captured conversation from the processor 108 before and/or after input (e.g., an audio file and/or an audio signal) to the processor 108 is transformed by the processor 108 into output (e.g., transformed conversation).

In some examples, the presenter 110 is or is part of a mobile device, a web browser, a computer, a watch, a phone, a tablet, a robot, a projector, a television, and/or a display. In some embodiments, the presenter 110 is provided similarly to the interface 104 by the same device. In some examples, a mobile phone is configured to provide both the interface 104 (e.g., touchscreen) and the presenter 110 (e.g., display). In certain examples, the interface 104 (e.g., touchscreen) of the mobile phone is configured to also function as the presenter 110 (e.g., display).

In certain embodiments, the presenter 110 includes a presenter interface configured for a user, an analyzer, and/or a recipient to interact with, edit, and/or manipulate the presented conversation. In some examples, the presenter 110 is communicatively coupled to the controller 102 such that the controller 102 provides instructions to the presenter 110, such as to switch the presenter 110 on (e.g., presenting a transformed conversation) and/or switch the presenter 110 off.

As discussed above and further emphasized here, FIG. 1 is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. In certain examples, the system 100 further includes other components and/or features in addition to the controller 102, the interface 104, the sensor 106, the processor 108, and/or the presenter 110. For example, the system 100 includes one or more sensors additional to the sensor 106, such as a camera, an accelerometer, a temperature sensor, a proximity sensor, a barometer, a biometric sensor, a gyroscope, a magnetometer, a light sensor, and/or a positioning system (e.g., a GPS).

FIG. 2 is a simplified diagram showing a method for processing and presenting one or more conversations according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The method 1000 includes process 1100 for receiving one or more instructions, process 1200 for capturing one or more conversations, process 1300 for auto-

matically transforming one or more conversations, and process 1400 for presenting one or more transformed conversations. Although the above has been shown using a selected group of processes for the method, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

In some examples, some or all processes (e.g., steps) of the method 1000 are performed by the system 100. In certain examples, some or all processes (e.g., steps) of the method 1000 are performed by a computer and/or a processor directed by a code. For example, a computer includes a server computer and/or a client computer (e.g., a smartphone). In some examples, some or all processes (e.g., steps) of the method 1000 are performed according to instructions included by a non-transitory computer-readable medium (e.g., in a computer program product, such as a mobile app and/or a web app). For example, a non-transitory computer-readable medium is readable by a computer including a server computer and/or a client computer (e.g., a smartphone). As an example, instructions included by a non-transitory computer-readable medium are executed by a processor including a processor of a server computer and/or a processor of a client computer (e.g., a smartphone).

At the process 1100, one or more instructions are received. In some examples, one or more instructions are provided by a user (e.g., a human, and/or a hardware and/or software system) and received by one or more components of the system 100 described above, such as received by the interface 104, the controller 102, the sensor 106, the processor 108, and/or the presenter 110. For example, the one or more instructions include a direct instruction (e.g., when the instruction is provided directly to a component) and/or an indirect instruction (e.g., when the instruction is provided to a gateway component which then instructs the component of interest to perform a process).

In certain examples, the one or more instructions cause the controller 102 to switch the sensor 106 between a capturing state and an idling state. For example, in the capturing state, the sensor 106 captures one or more conversations. In another example, in the idling state, the sensor 106 does not capture any conversation. In some examples, receiving a direct instruction includes a user directly switching on the sensor 106 to start the capturing of a conversation. In certain examples, receiving an indirect instruction includes receiving a start instruction via the interface 104, which then instructs the controller 102 to instruct the sensor 106 to start capturing a conversation.

At the process 1200, one or more conversations (e.g., one or more human-to-human conversations) are captured. In some examples, one or more conversations (e.g., a meeting conversation and/or a phone conversation) are captured by live recording via the sensor 106 (e.g., a microphone, a phone, a receiver, and/or a computing device). In certain examples, one or more conversations are captured by loading (e.g., by wire and/or wirelessly) one or more conversations in audio form (e.g., a .mp3 file, a .wav file, and/or a .m4a file). In some embodiments, capturing one or more conversations include capturing an incoming and/or outgoing phone conversation. In some embodiments, capturing one or more conversations includes capturing minutes, notes, ideas, and/or action items (e.g., of a meeting). In some embodiments, capturing one or more conversations includes capturing metadata corresponding to the one or more conversations, and the metadata include date of capture, time of

capture, duration of capture, and/or title of the capture (e.g., a title that is entered via the interface 104).

In some embodiments, capturing one or more conversations includes utilizing one or more components (e.g., the sensor 106, the controller 102, the processor 108, and/or the interface 104) of the system 100 and/or utilizing one or more components external to the system 100. In some examples, the sensor 106 of the system 100 is configured to capture a live conversation. In certain examples, the controller 102 and/or the processor 108 are configured to receive a pre-recorded conversation (e.g., a .mp3 file, a .wav file, and/or a .m4a file). In some examples, the interface 104 is configured to capture metadata associated to the conversation. In certain examples, a clock (e.g., of the system 100 or external to the system 100) is configured to provide date and time information associated to the conversation.

At the process 1300, one or more conversations (e.g., the one or more conversations captured at the process 1200) are transformed (e.g., transcribed, extracted, converted, summarized, and/or processed) automatically. In some examples, the captured conversations are transformed by the processor 108. In certain examples, the process 1300 is implemented according to FIG. 3.

FIG. 3 is a simplified diagram showing the process 1300 for automatically transforming one or more conversations according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The process 1300 includes process 1302 for receiving a conversation, process 1304 for automatically transcribing the conversation to synchronized text (e.g., synchronized transcript), process 1306 for automatically segmenting the conversation in audio form and the synchronized text, process 1308 for automatically assigning a speaker label to each conversation segment, and process 1310 for sending the transformed conversation (e.g., including synchronized text with speaker-labeled conversation segments). Although the above has been shown using a selected group of processes for the process 1300, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

In some examples, some or all processes (e.g., steps) of the process 1300 are performed by the system 100. In certain examples, some or all processes (e.g., steps) of the process 1300 are performed by a computer and/or a processor directed by a code. For example, a computer includes a server computer and/or a client computer (e.g., a smartphone). In some examples, some or all processes (e.g., steps) of the process 1300 are performed according to instructions included by a non-transitory computer-readable medium (e.g., in a computer program product, such as a mobile app and/or a web app). For example, a non-transitory computer-readable medium is readable by a computer including a server computer and/or a client computer (e.g., a smartphone). As an example, instructions included by a non-transitory computer-readable medium are executed by a processor including a processor of a server computer and/or a processor of a client computer (e.g., a smartphone).

At the process 1302, a conversation (e.g., a human-to-human conversation) is received. For example, a conversation is received by the system 100, such as by the processor 108. In some embodiments, the conversation (e.g., a human-to-human conversation) received in process 1302 is in audio

form (e.g., sound wave and/or digital signal) and is captured by and/or sent from the sensor **106** of the system **100**. In some embodiments, the conversation received in process **1302** is a live recording (e.g., a live recording of a human-to-human conversation). In some examples, the conversation is received (e.g., by the processor **108** of the system **100**) continuously and/or intermittently (e.g., via fixed frequency push). In certain examples, the conversation is received (e.g., by the processor **108** of the system **100**) in real-time and/or in near real-time (e.g., with a time delay less than 5 minutes, 1 minutes, or 4 seconds between capture and reception of a conversation).

In certain embodiments, the conversation (e.g., a human-to-human conversation) received in process **1302** is a pre-recorded conversation in audio form (e.g., sound wave and/or digital signal). For example, the pre-recorded conversation is an audio recording (e.g., a .mp3 file, a .wav file, and/or a .m4a file) uploaded from an internal device and/or an external device (e.g., a local storage device such as a hard drive, and/or a remote storage device such as cloud storage). In some examples, the conversation received in process **1302** is a phone conversation. In certain examples, the conversation is automatically received in process **1302**, such as by the processor **108**, such as whenever a conversation is sent to the processor **108** (e.g., from the sensor **106** and/or from the controller **102**).

At the process **1304**, a conversation (e.g., an audio-form conversation received at process **1302**) is automatically transcribed into synchronized text. In some embodiments, the conversation is automatically transcribed (e.g., with no user input or with minimal user input). In some examples, the transcribing is performed by at least the processor **108** of the system **100**. In certain examples, the transcribing is performed by the processor **108** and also modified by a human. In some embodiments, the conversation transcribed at process **1304** includes the conversation received at process **1302**, which is in audio form (e.g., sound wave and/or digital signal).

In some embodiments, the text (e.g., the transcript) generated at process **1304** includes English words, phrases, and/or terms. In certain embodiments, the audio-form conversation received at process **1302** and the text generated at process **1304** are timestamped and/or indexed with time, to synchronize the audio and the text. For example, the audio-form conversation received at process **1302** and the text (e.g., the transcript) generated at process **1304** are synchronized. In some examples, the text (e.g., the transcript) generated at process **1304** is searchable. For example, the text (e.g., the transcript) is searchable via a search bar. In certain examples, once transcribed at process **1304**, the conversation (e.g., from process **1302**) becomes a transcribed conversation including both audio and text that is synchronized with the audio.

At the process **1306**, a conversation in audio form (e.g., the conversation in audio form received at process **1302**) and a synchronized text (e.g., the synchronized text generated at process **1304**) are automatically segmented. In some embodiments, the audio-form conversation and the synchronized text are automatically segmented (e.g., with no user input or with minimal user input), and the segmented audio-form conversation and the segmented synchronized text are automatically generated. In some examples, the segmenting is performed by the processor **108** of the system **100**. In certain examples, the segmenting is performed by the processor **108** and also modified by a human. In certain embodiments, the conversation (e.g., audio-form conversation and/or the synchronized text) is segmented at process

1304 into different segments when a speaker change occurs and/or a natural pause occurs. In some embodiments, each segment of the audio-form conversation and the synchronized text generated at process **1306** is associated with one or more timestamps, each timestamp corresponding to the start time, and/or the end time. In certain embodiments, each segment of the audio-form conversation and the synchronized text generated at process **1306** is associated with a segment timestamp, the segment timestamp indicating the start time, the segment duration, and/or the end time.

In some embodiments, the audio-form conversation and the synchronized text are segmented at process **1306** into a plurality of segments that include one or more segments corresponding to the same speaker. In some examples, each segment is spoken by a single speaker. For example, the processor **140** is configured to automatically distinguish one or more speakers of the audio-form conversation. In certain examples, multiple segments spoken by the same speaker are next to each other and/or are separated by one or more segments spoken by one or more other speakers.

In certain embodiments, once segmented at process **1306**, the audio-form conversation (e.g., the conversation in audio form received at process **1302**) and the synchronized text (e.g., the synchronized text generated at process **1304**) becomes a segmented audio-form conversation and a segmented synchronized text. In some embodiments, segments of the audio-form conversation and segments of the synchronized text have one-to-one correspondence relationship. In some examples, each segment of audio-form conversation corresponds to one segment of synchronized text, and the segment of synchronized text is synchronized with that segment of audio-form conversation. In certain examples, different segments of audio-form conversation correspond to different segments of synchronized text, and the different segments of synchronized text are synchronized with the different segments of audio-form conversation respectively.

At the process **1308**, a speaker label is automatically assigned to each segment of text synchronized to one segment of audio-form conversation as generated by the process **1306**. In some embodiments, the speaker label is automatically assigned (e.g., with no user input or minimal user input), and the speaker-assigned segmented synchronized text and corresponding segmented audio-form conversation are automatically generated. In some examples, the assigning of speaker label is performed by the processor **108** of the system **100**. In certain examples, the assigning of speaker label is performed by the processor **108** and also modified by a human. In some embodiments, the speaker label includes a speaker name and/or a speaker picture.

In some embodiments, at the process **1308**, one or more segments of text, which are synchronized to one or more corresponding segments of audio-form conversation, are grouped into one or more segment sets each associated with the same speaker pending a speaker label assignment. In those embodiments, the speaker label is assigned to each segment set, which in turn assign the speaker label to all segments belonging to the segment set.

In some embodiments, at the process **1308**, the speaker label is assigned to each segment of text synchronized to one corresponding segment of audio-form conversation, by matching a voiceprint of the corresponding segment of audio-form conversation to a reference voiceprint corresponding to a speaker label.

In certain embodiments, the process **1308** includes assigning an "unknown" speaker label (e.g., with no name and/or with a placeholder picture) to a segment. In some embodiments, once assigned with one or more speaker labels at

11

process 1308, the segmented text that is synchronized with the segmented audio-form conversation (e.g., as generated at process 1306) becomes a speaker-assigned segmented text that is synchronized with the segmented audio-form conversation, with a speaker label assigned to each segment.

In some embodiments, a speaker corresponds to a speaker label, but a speaker label may or may not include a speaker name. In some examples, the speaker label corresponding to an unknown speaker does not include a speaker name. In certain examples, the process 1300 automatically identifies a new speaker voiceprint, but the user has not provided the name and/or the picture of the speaker yet; hence the speaker is determined to be, for example, an unknown speaker.

At the process 1310, a transformed conversation (e.g., including the speaker-assigned segmented synchronized text and its corresponding segmented audio-form conversation) is sent. For example, the transformed conversation is sent from the processor 108 to the controller 102 and/or to the presenter 110. In some embodiments, the transformed conversation sent at process 1310 includes the speaker-assigned segmented synchronized text and its corresponding segmented audio-form conversation as generated by the process 1308. In certain embodiments, the transformed conversation sent at process 1310 includes the segmented audio-form conversation and the segmented synchronized text as generated by the process 1306.

In some embodiments, the transformed conversation includes segmented audio, segmented text synchronized with segmented audio, speaker labels (e.g., name and/or picture) associated with the segments, and/or metadata (e.g., including a date, a time, a duration and/or a title). In certain embodiments, the transformed conversation is sent automatically, for example, by the processor 108. In certain embodiments, the transformed conversation is further sent or shared with other users, for example, via email.

As discussed above and further emphasized here, FIG. 3 is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. In some examples, the process 1304 and the process 1306 are modified such that segmenting the conversation in audio form occurs before synchronized text is transcribed for each segment. In certain examples, the process 1308, at which one or more speaker labels are assigned, occurs before transcribing the conversation in audio form and/or segmenting the conversation in audio form.

In certain embodiments, transcribing, segmenting, and/or assigning speaker label to a conversation are performed with the aid of a user and/or human. For example, a transcript automatically generated (e.g., at process 1304) is editable (e.g., by a user and/or human). In yet another example, segments automatically generated (e.g., at process 1306) is editable to split one segment and/or combine multiple segments (e.g., by a user and/or human). In yet another example, speaker labels automatically assigned (e.g., at process 1308) are editable (e.g., by a user and/or human).

In certain embodiments, the conversation to which transcribing, segmenting, and/or assigning speaker label are performed includes the conversation in audio form or the transcription. In some examples, the conversation in audio form is first segmented and/or speaker-assigned, and followed by having each segment transcribed to generate the synchronized text associated with each segment of conversation in audio form. In certain examples, the conversation in audio form is first transcribed to generate synchronized transcript, and followed by segmenting and/or assigning speaker label to the transcript. For example, the conversation

12

in audio form is not directly segmented, but instead is indirectly segmented or remains unsegmented and merely corresponds to the transcript in a word-by-word relationship (e.g., each transcribed text corresponds to a timestamp with an associated audio).

Returning to FIG. 2, at process 1400, one or more transformed conversations (e.g., the transformed conversation sent at the process 1310) are presented. In certain embodiments, the process 1400 includes presenting the transformed conversation (e.g., including the speaker-assigned segmented synchronized text and its corresponding segmented audio-form conversation) with the presenter 110. In some examples, when the audio-form conversation is played, the corresponding word in the synchronized text is highlighted when the word is spoken. In certain examples, the text is synchronized with the audio-form conversation at both the segment level and the word level.

In certain embodiments, the process 1400 includes presenting the metadata associated with the transformed conversation. For example, the metadata include a date (e.g., of capturing, processing, or presenting), a time (e.g., of capturing, processing, or presenting), a duration (e.g., of the conversation), and/or a title. In some embodiments, the process 1400 includes presenting a player, such as an audio player. For example, the audio player is a navigable audio player configured to provide control (e.g., to a user) such that the presenting of the transformed conversation is interactive.

In some embodiments, the process 1400 includes presenting the speaker-assigned segmented synchronized text (e.g., generated by the process 1308) in a searchable manner, such as via a search bar. In some embodiments, the process 1400 includes presenting search results that match a searched text (e.g., via the search bar) in the speaker-assigned segmented synchronized text in a first marked form, such as a first highlighted form (e.g., highlighted in saturated and/or faded yellow).

In certain embodiments, at the process 1400, the transformed conversation is presented such that the search results (e.g., in the speaker-assigned segmented synchronized text) and/or the audio corresponding to the search results (e.g., indexed with the same timestamp) are highlighted, such as in a first marked form. In some embodiments, the text being presented (e.g., matching the audio during a playback or when paused) is highlighted, such as in a second marked form, (e.g., highlighted in green). For example, the text being presented (e.g., the text being played back) is indexed with the same timestamp as the audio instance within the conversation, such as at a particular time indicated by a progress indicator along a progress bar.

As discussed above and further emphasized here, FIG. 1, FIG. 2, and FIG. 3 are merely examples, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. In some examples, the system as shown in FIG. 1 is used to process and present a speech by a single-speaker and/or a conversation made by a single speaker talking to himself or herself. In certain examples, the method as shown in FIG. 1 is used to process and present a speech by a single-speaker and/or a conversation made by a single speaker talking to himself or herself. In some examples, the process 1300 as shown in FIG. 3 is used to automatically transform a speech by a single-speaker and/or a conversation made by a single speaker talking to himself or herself.

According to certain embodiments, a system is configured to capture, process, render, and/or display one or more

13

context-aware moment-associating elements (e.g., one or more speeches and/or one or more photos). For example, the system is described and/or implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15. In one embodiment, the context includes location, time, one or more participants, topic, historical information, and/or a person's social network profile and/or history. In another embodiment, the context is used by the system such that, based on the context, the interpretation of the current speech and/or the current conversation can be different. For example, the historical information is used by the system such that, based on what a person spoke or heard in the past, the interpretation of the current speech and/or the current conversation can be different.

FIG. 4 is a simplified diagram showing a method of capturing and displaying one or more moments (e.g., one or more multi-party speeches, and/or one or more inline photos) according to certain embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The method 4000 includes processes 4002, 4004, 4006, 4008, and 4010. In some examples, the method 4000 is performed by the system 100. In certain examples, the method 4000 is the same as the method 1000. For example, the one or more moments include one or more multi-party speeches, and/or one or more inline photos.

Although the above has been shown using a selected group of processes for the method 4000, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

At process 4002, capturing one or more moment-associating elements using one or more moment-capturing devices is performed. For example, one or more moment-associating elements are captured using one or more moment-capturing devices (e.g., the sensor 106). At process 4004, transmitting the captured one or more moment-associating elements to a processor is performed. For example, the captured one or more moment-associating elements are transmitted to a processor (e.g., the processor 108).

At process 4006, transforming the one or more moment-associating elements into one or more moment-associating information using the processor. For example, the one or more moment-associating elements are transformed into one or more moment-associating information using the processor (e.g., the processor 108). At process 4008, transmitting at least one of the one or more moment-associating information to one or more moment-displaying devices and/or the moment-capturing devices is performed. For example, at least one of the one or more moment-associating information is transmitted to one or more moment-displaying devices (e.g., the presenter 110) and/or the moment-capturing devices. At process 4010, displaying at least one of the moment-associating information is performed. For example, at least one of the moment-associating information is displayed (e.g., by the presenter 110).

FIG. 5 is a simplified diagram showing process 4006 for transforming the one or more moment-associating elements into one or more moment-associating information according to some embodiments of the present invention. The process 4006 of the method 4000 includes processes 5002, 5004, 5006, 5008, 5010, and 5012. In some examples, the method

14

4006 is performed by the processor 108. In certain examples, the process 4006 is the same as the process 1300.

Although the above has been shown using a selected group of processes for the process 4006, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

At process 5002, segmenting speech audio elements into one or more speech audio segments using speaker change detection is performed. At process 5004, identifying and assigning a segment speaker for each audio segment is performed. At process 5006, transcribing speech audio elements into segmented and speaker-identified transcription is performed. At process 5008, generating capitalized, punctuated, and segmented transcription with one or more time-stamps using automated speech recognition is performed. At process 5010, generating key phrases, action items, summary, statistics, and/or analytics is performed. At process 5012, encoding segmented and speaker-identified transcription and/or the speech audio elements into compressed format and/or format for playback, streaming, or editing. For example, segmented and speaker-identified transcription and/or the speech audio elements are encoded into compressed format and/or format for playback, streaming, and/or editing.

FIG. 6 is a simplified diagram showing a system for capturing, processing, and rendering a context-aware moment-associating element according to certain embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The system 600 includes a capturing device 602, a local storage 612, an application programming interface (API) 604, an automatic speech recognition (ASR) system 614, a key-value database 606, a dynamic server 608, and/or a displaying device 610. In some examples, the system 600 is the same as the system 100. In certain examples, the system 600 performs the method 1000 and/or the method 4000. For example, the context-aware moment-associating element includes a multi-party speech and/or an inline photo.

Although the above has been shown using a selected group of components for the system 600, there can be many alternatives, modifications, and variations. For example, some of the components may be expanded and/or combined. Other components may be inserted to those noted above. Depending upon the embodiment, the arrangement of components may be interchanged with others replaced. Further details of these components are found throughout the present specification.

As an example, the representational state transfer (REST) API is removed from the system 600 as shown in FIG. 6. For example, the real time API is removed from the system 600 as shown in FIG. 6. As an example, both the real time API and the representational state transfer (REST) API are removed from the system 600 as shown in FIG. 6, and are replaced by one or more other protocols.

In some embodiments, the component 602 is a capturing device such as an App (e.g., on IOS, Android, or ChromeOS) or a Browser (e.g., java-based). In certain embodiments, the component 604 is an application programming interface (API). In some embodiments, the component 606 is a key-value (K-V) database (e.g., a database that stores time sequence and audio recording). In certain embodiments, the

component **608** is a dynamic server (e.g., Amazon Web Services). For example, the dynamic server **608** stores one or more dynamic libraries.

In some embodiments, the component **610** is a displaying device (e.g., for playback, streaming, and/or editing). In certain embodiments, the component **612** is a local storage. In some embodiments, the component **614** is an automatic speech recognition (ASR) system for transcribing the audio recording into information (e.g., start time and end time of a phrase, and/or start time and end time of text).

As discussed above and further emphasized here, FIG. **6** is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, the system **600** is described and/or implemented according to at least FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4**, FIG. **5**, FIG. **6**, FIG. **7**, FIG. **8**, FIG. **9**, FIG. **10**, FIG. **11**, FIG. **12**, FIG. **13**, FIG. **14**, and/or FIG. **15**.

FIG. **7** is a simplified diagram showing a method of operation for the system **600** as shown in FIG. **6** according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The method **7000** includes processes **7002**, **7004**, **7006**, and **7008**. For example, the method **7000** is the same as the method **1000** and/or the method **4000**. As an example, the method **7000** is performed by the system **100** and/or the system **600**. Although the above has been shown using a selected group of processes for the method **7000**, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

At process **7002**, recording one or more speech audios, one or more videos and/or one or more pictures in real time on one or more mobile phones and/or one or more browsers and/or importing from one or more other sources are performed. For example, the one or more pictures are one or more photos. At process **7004**, processing the one or more speech audios, the one or more videos, and/or one or more pictures, and/or generating one or more transcripts in real-time are performed.

At process **7006**, incrementally identifying one or more speakers, segmenting one or more audios into one or more bubbles, reprocessing one or more entire meetings, and/or encoding one or more audios are performed. At process **7008**, pushing the processed one or more audios, one or more videos, one or more pictures, and/or one or more transcripts to one or more clients and/or presenting to one or more users are performed.

FIG. **8** is a simplified diagram showing a process of operation for front-end and/or client related to FIG. **7** according to certain embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The process **8000** includes processes **8002** and **8004**. For example, the process **8000** is related to at least the process **7002** as shown in FIG. **7**. Although the above has been shown using a selected group of processes for the process **8000**, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted

to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

At process **8002**, capturing one or more speeches and saving to one or more local disks are performed. At process **8004**, establishing one or more persistent connections with one or more servers to upload one or more audios and/or getting one or more transcript updates in real time are performed.

FIG. **9** is a simplified diagram showing a process of operation for backend incremental processing related to FIG. **7** according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The process **9000** includes processes **9002**, **9004**, **9006**, and **9008**. For example, the process **9000** is related to at least the process **7006** as shown in FIG. **7**. Although the above has been shown using a selected group of processes for the process **9000**, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

At process **9002**, segmenting one or more audios based on one or more speaker change detections is performed. At process **9004**, identifying one or more speakers for each audio segment is performed. At process **9006**, reprocessing one or more entire transcripts based on segmentation and speaker identification (ID), and/or generating one or more key phrases, one or more action items, one or more summaries, one or more statistics, and/or one or more analytics are performed. At process **9008**, encoding one or more audios into one or more compressed formats and/or streaming formats for one or more playbacks, and/or processing one or more images and/or one or more videos are performed.

FIG. **10** is a simplified diagram showing a process of operation for backend real-time processing related to FIG. **7** according to certain embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The process **10000** includes processes **10002**, **10004**, **10006**, and **10008**. For example, the process **10000** is related to at least the process **7004** as shown in FIG. **7**. Although the above has been shown using a selected group of processes for the process **10000**, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

At process **10002**, feeding one or more audios into one or more automated speech recognition stream servers, and/or generating one or more transcript words with one or more timestamps for one or more current windows are performed. At process **10004**, generating one or more capitalizations and/or one or more punctuations, and/or segmenting one or more audios and/or one or more transcripts into one or more segments are performed.

At process **10006**, saving one or more audios, one or more videos, one or more pictures, and/or one or more transcripts into one or more persistent storages, assigning one or more group sharing permissions, and/or performing one or more minute accountings for one or more payment statistics are

17

performed. At process **10008**, pushing one or more transcripts to one or more clients via one or more persistent network connections is performed.

FIG. **11** is a simplified diagram showing a method for real-time capturing of one or more inline photos according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The method **11000** includes processes **11002**, **11004**, **11006**, and **11008**. For example, the method **11000** is performed with one or more long-form, multi-party, and/or far-field conversations with voice, photo, and/or video. As an example, the method **11000** is one or more parts of the method **1000**, one or more parts of the method **4000**, and/or one or more parts of the method **7000**.

Although the above has been shown using a selected group of processes for the method **11000**, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

At process **11002**, a user takes at least one photo while recording using an App (e.g., using the Otter App). At process **11004**, the App syncs the photo along with the timestamp to a server. At process **11006**, the server stores the photo and timestamp and associates them to the transcript of the conversation being recorded. At process **11008**, the App (e.g., Otter App) inserts the photo inline with the real-time transcript for display to the user. For example, the photo is an inline photo.

FIG. **12** is a simplified diagram showing a method for rendering of one or more inline photos according to certain embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The method **12000** includes processes **12002**, **12004**, **12006**, **12008**, **12010**, and **12012**. For example, the method **12000** is performed with one or more long-form, multi-party, and/or far-field conversations with voice, photo, and/or video. As an example, the method **12000** is one or more parts of the method **1000**, one or more parts of the method **4000**, and/or one or more parts of the method **7000**.

At process **12002**, a user opens a conversation in an App (e.g., in the Otter App). At process **12004**, the App (e.g., the Otter App) requests a server for one or more conversation details. At process **12006**, the server sends back data of one or more conversation details, including multiple resolutions of each inline photo. For example, the multiple resolutions include a low resolution, a medium resolution, and a high resolution.

At process **12008**, the App (e.g., the Otter App) renders a row of low-resolution thumbnails at the top of the conversation detail view. For example, different thumbnails of the low-resolution thumbnails correspond to different inline photos respectively. At process **12010**, the App (e.g., the Otter App) renders a medium-resolution version of each photo inline with the transcript based on timestamp. At process **12012**, the App (e.g., the Otter App) renders a high-resolution version of each inline photo for full-screen gallery view.

FIG. **13** is a simplified diagram showing a system for real-time capture, processing, and rendering of one or more context-aware conversations according to some embodiments of the present invention. This diagram is merely an

18

example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The system **13000** includes one or more Apps, such as an Android App **13002**, an iOS App **13004**, and/or a web App (e.g., otter.ai). Additionally, the system **13000** includes N application programming interface (API) servers, with N being a positive integer larger than or equal to 1. Moreover, the system **13000** includes N automated speech recognition (ASR) systems, with N being a positive integer larger than or equal to 1. In some examples, the system **13000** is the same as the system **100**. In certain examples, the system **13000** is the same as the system **600**. In some examples, the system **13000**, the system **100**, and the system **600** are the same. For example, the system **13000** performs the method **1000**, the method **4000** and/or the method **7000**. As an example, the system **13000** performs the process **10000** and/or the process **11000**.

Although the above has been shown using a selected group of components for the system **13000**, there can be many alternatives, modifications, and variations. For example, some of the components may be expanded and/or combined. Other components may be inserted to those noted above. Depending upon the embodiment, the arrangement of components may be interchanged with others replaced. Further details of these components are found throughout the present specification.

In some embodiments, the one or more conversations that are real-time captured, processed, and rendered by the system **13000** include one or more long-form, multi-party, and/or far-field conversations with voice, photo, and/or video. In certain embodiments, the one or more Apps, such as an Android App **13002**, an iOS App **13004**, and/or a web App (e.g., otter.ai) are configured to perform capturing and/or rendering. For example, each web app is a frame and/or a widget. As an example, the one or more Apps are configured to send information to the N application programming interface (API) servers to sync to cloud.

In some embodiments, the N application programming interface (API) servers and the N automated speech recognition (ASR) systems are configured to perform transcribing and/or extracting. For example, the N application programming interface (API) servers are configured to perform speaker diarization, identification, and/or punctuation. As an example, the N automated speech recognition (ASR) systems are configured to receive information from the N application programming interface (API) servers. For example, the N automated speech recognition (ASR) systems are configured to use at least one acoustic model (AM) and/or at least one language mode (LM).

As discussed above and further emphasized here, FIG. **13** is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, "Web App" (e.g., a frame and/or a widget) can be embedded in "otter.ai". In yet another example, "otter.ai" is removed from FIG. **13**, and "Web App" (e.g., a frame and/or a widget) is embedded in a different internet website and/or in a mobile app.

FIG. **14** is a simplified diagram showing a method of training the system **13000** as shown in FIG. **13** according to certain embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The method **14000** includes processes **14002**, **14004**, and **14006**. For example, the method **14000** is implemented to

train the system for real-time capture, processing, and rendering of one or more long-form, multi-party, and/or far-field conversations with voice, photo, and/or video. Although the above has been shown using a selected group of processes for the method **14000**, there can be many alternatives, modifications, and variations. For example, some of the processes may be expanded and/or combined. Other processes may be inserted to those noted above. Depending upon the embodiment, the sequence of processes may be interchanged with others replaced.

At process **14002**, training data are provided. For example, the training data include one or more human speech audios and/or one or more corresponding transcripts from public domain. As an example, the training data include one or more accented speeches. For examples, the training data including data from one or more meetings, one or more conferences, and/or one or more calls. As an example, training data are provided with one or more speaker names labeled. At process **14004**, model training is performed. For example, the model training is based on deep neural network (DNN). At process **14006**, one or more models are provided. For example, the models include one or more acoustic models (AMs), one or more language models (LMs), and/or one or more speaker identification (ID) models.

FIG. **15** is a simplified diagram showing an example of user interface for one or more inline photos related to the method **11000** as shown in FIG. **11** and/or the method **12000** as shown in FIG. **12** according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications.

As shown in FIG. **6**, the system **600** comprises the moment-capturing device **602**, the application programming interface (API) **604**, the automatic speech recognition (ASR) system **614**, the key-value database **606**, one or more dynamic libraries (e.g., related to the dynamic server **608**), and optionally the moment-displaying device **610** according to some embodiments. In some examples, the ASR system **614** includes one or more software modules and/or one or more processors. For example, the ASR system **614** is a software system. As an example, the ASR system **614** is a hardware system. For example, the ASR system **614** is a software and hardware system.

As shown in FIG. **4**, the moment-capturing device **602** is configured to detect, sense, capture, record, and/or analyze one or more moment-associating elements according to various embodiments. For example, the one or more moment-associating elements include one or more audio elements, one or more visual elements, and/or one or more environmental elements. In another example, it is to be appreciated that one or more moment-capturing devices **602** are utilized in the system **600** and each device captures one or more moment-associating elements characterizing the moment where the elements are captured. In some examples, one or more moments are captured by the system **600**, and each capture (e.g., an audio element) is assigned a timestamp for the moment-associating elements to be associated to. For example, a first audio element is associated with a first timestamp indicating that the first audio element is captured at a first moment in which the first timestamp corresponds to.

In some embodiments, the one or more audio elements include one or more voice elements of one or more voice-generating sources (e.g., a user and/or a voice-generating

device) and/or one or more ambient sound elements (e.g., sound elements from traffic, music, and/or nature).

In some embodiments, the one or more visual elements include one or more pictures, one or more images, one or more screenshots, one or more video frames, one or more projections, and/or one or more holograms, each corresponding to a timestamp associated with the moment in which the visual element(s) is captured.

In some embodiments, the one or more environmental elements include a global position (e.g., longitude, latitude, altitude, country, city, street), a location type (e.g., home, office, school, coffee shop, indoor, outdoor), a moment condition (e.g., temperature, humidity, movement, velocity, direction, ambient noise level, echo properties).

According to some embodiments, the moment-capturing device **602** includes a stationary device (e.g., a computer, a television, and/or a home appliance) and/or a portable device (e.g., a laptop, a phone, a tablet, a watch, a pair of glasses, apparel, a pair of shoes, and/or an accessory).

In various examples, the moment-capturing device **602** is also a moment-displaying device and/or including a local storage configured to store moment-associating elements and/or its processed form (e.g., transcription). As shown in FIG. **8**, the system is configured to establish persistent connection with a server to upload audio and receive transcript update in real time, according to some embodiments.

As shown in FIG. **11**, in various examples, a moment-capturing device (e.g., the moment-capturing device **602**) includes one or more element-capturing device (e.g., one or more cameras, one or more microphones, one or more touchscreens, one or more global positioning systems, one or more thermometers, one or more compasses, one or more photo sensors, one or more accelerometers, one or more pedometers, one or more heart rate sensors, one or more humidity sensors, one or more pressure sensors, and/or one or more wear sensors) configured to capture one or more moment-associating elements (e.g., picture, video, sound, speech, voice, touch, gesture, position, location, environmental setting, direction, movement, brightness, time-of-day, velocity, distance-from-reference, heart rate, humidity, pressure, and/or degree of wear). According to certain embodiments, each of the moment-associating elements captured by the one or more element-capturing devices is coupled to a timestamp representing and/or corresponding to the time of capture, and is stored in a local storage (e.g., the local storage **612**), a dynamic storage (e.g., related to the key-value database **606** and/or the dynamic server **608**), and/or a web server storage, as shown in FIG. **10**.

In some embodiments, one or more of the moment-associating elements captured are processed and/or transformed into moment-corresponding information (e.g., text) which represents the one or more of the moment-associating elements (e.g., by the system **600** according to the method **7000**). For example, voice captured from a speech (i.e., a moment-associating element) is transcribed (i.e., processed) into text (i.e., a moment-associating information). In some examples, a sequence of moment-associating elements are processed (e.g., in conjunction) such that additional moment-associating information is extrapolated from processing. For example, processing a single word recorded is not able to indicate the tone of how the word is spoken. However, processing a sentence including the word as well as additional words captured at different moments enables a corresponding tone to be extrapolated, according to some embodiments.

In some embodiments, as shown in FIG. **6**, the application programming interface (API) **604** includes a real time API,

and/or a representational state transfer (REST) API. For example, the API 604 is configured to push and pull data (e.g., captured elements and/or transcription) at least between two of the moment-capturing device 602, the ASR system 614, the dynamic storage (e.g., related to the key-value database 606 and/or the dynamic server 608), and the moment-displaying device 610. In some embodiments, the frequency of the push and/or pull is designed, set, adjusted such that the moment-displaying device 610 displays a moment-associating information (e.g., transcription) substantially in real time with the capturing of its corresponding moment-associating element(s) (e.g., voice). For example, the frequency is designed such that transcription of a phrase appears on a mobile device (e.g., the moment-displaying device 610) in less than 5 minutes, such as 1 minute, such as 30 seconds, such as 5 seconds, such as 1 second, from the time of recording of the phrase by a mobile device (e.g., the moment-capturing device 602). In some examples, a device (e.g., a phone) is both the moment-capturing device 602 and the moment-displaying device 610.

In some embodiments, the one or more dynamic storage include a first storage (e.g., the key-value database 606) and a second storage (e.g., the dynamic server 608 such as a web storage server). For example, in the second storage (e.g., the web storage server), the original data (e.g., stored at the local storage 612 of the moment-capturing device 602) are processed (e.g., by the moment-capturing device 602) such that a reduced form of the data is transmitted to the first storage (e.g., the key-value database 606). In some examples, the reduced form of the data also include analytical information such as one or more timestamps. In some embodiments, the data in the reduced form are then processed (e.g., by the ASR system 614) to transform the moment-associating elements (e.g., audio) into moment-associating information (e.g., transcription) such that the processed, complex data are transmitted to the second storage (e.g., the dynamic server 608 such as a web storage server). For example, the data stored in the second storage are pulled by the moment-displaying device 610 for playback, streaming, and/or editing.

According to some embodiments, the ASR system 614 includes a model (e.g., a mathematical model) configured to receive an input of audio (e.g., speech, voice, and/or playback) and generates an output including audio-representing data such as transcription. For example, the output further includes information (e.g., timestamp, tone, volume, speaker identification, noise level, and/or background acoustic environment identification) associated with the transcription (e.g., associated with each sentence and/or associated with each phrase). In some examples, the model of the ASR system 614 is updated and/or improved, for example, by feeding training data to the model. In some examples, an improved model improves the accuracy and/or speed of the ASR system 614 in transcribing audio data.

As shown in FIG. 14, examples of training data include data including audio data and optionally corresponding transcription according to certain embodiments. For example, the corresponding transcription includes speaker identification (e.g., name, gender, age, and/or other elements that may correspond to the audio source such as a speaker), and/or environmental setting (e.g., indoor, outdoor, closed room, and/or open space). In some examples, the training data include one or more characterizing parameters including language spoken, speaker accent, and/or speech velocity. In some embodiments, feeding training data of a particular type improves capability (e.g., speed and/or accuracy) of the ASR system 614 to transcribe that particular type and

optionally similar type of audio data. For example, the more training data with a “business meeting” environmental setting that are fed to the model, the better the ASR system 614 is at transcribing new audio data (e.g., real time recordings) having a “business meeting” environmental setting.

According to some embodiments, the ASR system 614 (e.g., a model, such as a mathematical model, of the ASR system 614) is also updated and/or improved by other means such as user input and/or correction (e.g., via a user interface). For example, the ASR system 614 (e.g., a model, such as a mathematical model, of the ASR system 614) is configured to read a set of parameters (e.g., a user voiceprint) which improves the accuracy (e.g., by 10%, by 20%, by 50%, by 90% or more) and/or transcribing certain audio data (e.g., recording of the user’s speech, voice, and/or conversation). For example, the voiceprint of a user becomes more comprehensive as more audio recordings of the user are processed by the ASR system 614 (e.g., refining and/or storing audio cues such as waveform). In some examples, the audio data include short-form (e.g., command) audio data and/or long-form (e.g., conversation) audio data. In some examples, a model (e.g., a mathematical model) of the ASR system 614 includes sub-models such as an acoustic model (AM) and/or a language model (LM), each configured to help the ASR system 614 to recognize specific type of sound (e.g., human-speech sound, ambient sound, and/or environmental sound).

As shown in FIG. 5, a model (e.g., a mathematical model) of the ASR system 614 is improved in real time while audio data is transcribed according to some embodiments. In some embodiments, the moment-capturing device 602 (e.g., a microphone) captures audio data (e.g., a conversation) containing a plurality of segments including a first, a second, and a third segment. For example, a first speaker has said the first and third segment and the second segment spoken by a second speaker. In some examples, as shown in FIG. 7, the real-time transcribing system 600 utilizing the ASR system 614 may have mistakenly identified (e.g., incrementally identified) the speaker of the first segment (e.g., of a live recording and/or of an imported file) to be the second speaker and/or have incorrectly transcribed one or more words in the first sentence. In some examples, as the ASR system 614 proceeds to transcribe the second sentence, the increased amount of data fed to the ASR system 614, for example, helps the ASR system 614 to identify that the first sentence is spoken by someone different than the speaker who spoke the second sentence, and to automatically update the transcription of the first sentence (e.g., on the local storage, on the dynamic storage, and/or on the moment-displaying device) to correct its prior error. And further as the ASR system 614 proceeds to transcribe the third segment, the ASR system 614 improves the accuracy in transcribing speech of the first speaker, for example, owing to the increased amount of data fed through the ASR system 614 according to some embodiments.

According to certain embodiments, the ASR system 614 is also configured to segment an audio recording into segments, such as segments having the same speaker or speakers. For example, in addition to recognizing words, phrases, and other speech-characterizing characteristics (e.g., accent, tone, punctuation, volume, and/or speed), the ASR system 614 is configured to extrapolate a start time and an end time of each word, phrase, sentence, topic, and/or the times where speaker-change occurs. As shown in FIG. 9, using one or more of such information, the ASR system 614 segments an audio recording of a conversation into segments where any of such information changes (e.g., where a

speaker-change occurs) according to some embodiments. For example, timestamps are tagged, marked, and/or noted by the ASR system 614 to correspond such transition of information (e.g., a speaker change and/or a tone change) to a specific time of the speech. In some examples, only the timestamps of such transition points are marked with information change, or alternatively, all timestamps are marked by the information.

In various examples, one or more moment-capturing elements (e.g., a word, a phrase, a picture, a screenshot) are used as anchor points by the system 600 to enable a user to navigate the processed data (e.g., transcription) and/or the unprocessed data (e.g., audio recording), and/or to search (e.g., to keyword search) with improved usability according to some embodiments. For example, a transcription of a speech is navigable and/or searchable (e.g., by a user) to quickly (e.g., in less than a second) find where a keyword is said during the speech, and be able to be directed to that segment of the speech (i.e., unprocessed form) and/or to that segment of the transcription (i.e., processed data form). In some examples, the transcription additionally or alternatively includes one or more images (e.g., thumbnails) as anchor points, in which each of the images corresponds to a timestamp and thus the segment where such image is presented is associated to that particular moment or moments. In some embodiments, the images are presented in-line with the rest of the transcription. In some examples, images are presented and/or recreated separately from the text (e.g., a photo-gallery). For example, the one or more images are captured manually (e.g., by a user), and/or automatically (e.g., a computer capturing each slide of a slide show presentation at the time when the slide starts).

In some embodiments, one or more object and/or image recognition systems are utilized such that a user of the system 600 can navigate and/or search the data (e.g., the processed data and/or the unprocessed data) using information beyond text. For example, a user does not need to scroll through a gallery of more than 100 images to find the image of interest, instead, a user can input an image in which the system is configured to search its image anchor points and find the moment where the image inputted by the user matches the image anchor point the most (e.g., more than 60% match, more than 80% match, or more than 90% match).

As shown in FIG. 12 and/or FIG. 13, in some embodiments, the system 600 is a smart note-taking and/or collaboration application according to certain embodiments. For example, the system help aid focus, collaboration and/or efficiency in meetings, interviews, lectures, and/or other important conversations. In some examples, the system 600 ambiently records one or more conversations and transcribes the one or more conversations in real time and/or in near-real time (e.g., delayed by less than 1 min, less than 30 second, or less than 10 second). In some examples, with the power of artificial intelligence (AI), the system 600 automatically identifies keywords and/or speakers, and/or automatically segments an audio recording. In some examples, the system 600 enables the content to be searchable (e.g., via keyword search) and/or shareable (e.g., via social media).

As shown in FIG. 15, in some embodiments, the system 600 supports inline photos such that a user can input a visual (e.g., snap a photo of a whiteboard, a presenter, and/or screenshot) and instantly add the visual inline to the transcribed text corresponding to the audio being captured. This adds a new visual element to the conversation.

In some embodiments, the system (e.g., the system described and/or implemented according to at least FIG. 1,

FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15) supports call recording such that a user can capture audio without needing to activate a microphone manually (e.g., without needing to click a microphone button on a phone), instead, the system can be prompted to start capturing/recording when a call starts. For example, such feature enables the system to be utilized for sales calls, phone interviews, and/or other important conversations.

In some embodiments, the system (e.g., the system described and/or implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15) supports advanced export options such that a user can export processed and/or unprocessed data (e.g., audio, transcription, image, video) including one or more of speaker labels, one or more timestamps, one or more text formats (e.g., .txt, .pdf, and/or .vtt) and/or one or more audio formats (e.g., .mp3, m4a, and/or .wma).

In some embodiments, the system (e.g., the system described and/or implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15) is configured to export text (e.g., export text to a clipboard, and/or export text directly into one or more other applications and/or software) and/or to export audio (e.g., export audio in .mp3).

In some embodiments, the system (e.g., the system described and/or implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15) generates one or more notifications for notifying (e.g., for notifying a user). For example, a notification is a reminder to a user to start capturing of audio and/or one or more other moment-associating elements. For example, a notification is sent to a user at the start of a meeting recorded on a calendar (e.g., the start of a meeting recorded on the calendar that is synced to the system). In some examples, a notification is sent to a second user when a first user shares and/or sends a file to the second user.

In some embodiments, the system (e.g., the system described and/or implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15) is configured to rematch one or more speakers to identify and label the speakers automatically, after manually tagging a few voice samples of each speaker in a conversation.

In some embodiments, the system (e.g., the system described and/or implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15) is configured such that a user manages and/or edits the local storage (e.g., the local storage 612) such that the user can control how much space the system uses to store data (e.g., one or more local copies of the audio recordings).

In some embodiments, the system (e.g., the system described and/or implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15) supports one or more Bluetooth devices such that the one or more Bluetooth-connected devices (e.g., one or more Bluetooth headphones) can be used to capture moment-associating elements (e.g., voice).

In some embodiments, the system (e.g., the system described and/or implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG.

25

9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15) extracts keywords from an audio recording.

In some embodiments, the method (e.g., the method described and/or implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15) is performed automatically with respect to at least one process of the method. For example, some processes of the method are performed automatically. As an example, all processes of the method are performed automatically.

According to one embodiment, a computer-implemented method for receiving and processing one or more moment-associating elements includes receiving the one or more moment-associating elements, transforming the one or more moment-associating elements into one or more pieces of moment-associating information, and transmitting at least one piece of the one or more pieces of moment-associating information. The transforming the one or more moment-associating elements into one or more pieces of moment-associating information includes segmenting the one or more moment-associating elements into a plurality of moment-associating segments, assigning a segment speaker for each segment of the plurality of moment-associating segments, transcribing the plurality of moment-associating segments into a plurality of transcribed segments, and generating the one or more pieces of moment-associating information based on at least the plurality of transcribed segments and the segment speaker assigned for each segment of the plurality of moment-associating segments. For example, the computer-implemented method is implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15.

In some examples, the receiving the one or more moment-associating elements includes assigning a timestamp associated with each element of the one or more moment-associating elements. In certain examples, the receiving the one or more moment-associating elements includes at least one selected from a group consisting of receiving one or more audio elements, receiving one or more visual elements, and receiving one or more environmental elements. In some examples, the receiving one or more audio elements includes at least one selected from a group consisting of receiving one or more voice elements of one or more voice-generating sources and receiving one or more ambient sound elements.

In certain examples, the receiving one or more visual elements includes at least one selected from a group consisting of receiving one or more pictures, receiving one or more images, receiving one or more screenshots, receiving one or more video frames, receiving one or more projections, and receiving one or more holograms. In some examples, the receiving one or more environmental elements includes at least one selected from a group consisting of receiving one or more global positions, receiving one or more location types, and receiving one or more moment conditions. In certain examples, the receiving one or more environmental elements includes at least one selected from a group consisting of receiving a longitude, receiving a latitude, receiving an altitude, receiving a country, receiving a city, receiving a street, receiving a location type, receiving a temperature, receiving a humidity, receiving a movement, receiving a velocity of a movement, receiving a direction of a movement, receiving an ambient noise level, and receiving one or more echo properties.

In some examples, the transforming the one or more moment-associating elements into one or more pieces of

26

moment-associating information includes: segmenting the one or more audio elements into a plurality of audio segments; assigning a segment speaker for each segment of the plurality of audio segments; transcribing the plurality of audio segments into a plurality of text segments; and generating the one or more pieces of moment-associating information based on at least the plurality of text segments and the segment speaker assigned for each segment of the plurality of audio segments. In certain examples, the transcribing the plurality of audio segments into a plurality of text segments includes transcribing two or more segments of the plurality of audio segments in conjunction with each other.

In some examples, the computer-implemented method further includes receiving one or more voice elements of one or more voice-generating sources, and receiving one or more voiceprints corresponding to the one or more voice-generating sources respectively. In certain examples, the transforming the one or more moment-associating elements into one or more pieces of moment-associating information further includes at least one selected from a group consisting of: segmenting the one or more moment-associating elements into the plurality of moment-associating segments based on at least the one or more voiceprints; assigning the segment speaker for the each segment of the plurality of moment-associating segments based on at least the one or more voiceprints; and transcribing the plurality of moment-associating segments into the plurality of transcribed segments based on at least the one or more voiceprints. In some examples, the receiving one or more voiceprints corresponding to the one or more voice-generating sources respectively includes at least one selected from a group consisting of receiving one or more acoustic models corresponding to the one or more voice-generating sources respectively, and receiving one or more language models corresponding to the one or more voice-generating sources respectively.

In certain examples, the transcribing the plurality of moment-associating segments into a plurality of transcribed segments includes, transcribing a first segment of the plurality of moment-associating segments into a first transcribed segment of the plurality of transcribed segments, transcribing a second segment of the plurality of moment-associating segments into a second transcribed segment of the plurality of transcribed segments, and correcting the first transcribed segment based on at least the second transcribed segment. In some examples, the segmenting the one or more moment-associating elements into a plurality of moment-associating segments includes at least one selected from a group consisting of: determining one or more speaker-change timestamps, each timestamp of the one or more speaker-change timestamps corresponding to a timestamp when a speaker change occurs; determining one or more sentence-change timestamps, each timestamp of the one or more sentence-change timestamps corresponding to a timestamp when a sentence change occurs; and determining one or more topic-change timestamps, each timestamp of the one or more topic-change timestamps corresponding to a timestamp when a topic change occurs. In certain examples, the segmenting the one or more moment-associating elements into a plurality of moment-associating segments is performed based on at least one selected from a group consisting of: the one or more speaker-change timestamps; the one or more sentence-change timestamps; and the one or more topic-change timestamps.

In some examples, the computer-implemented method further includes establishing one or more anchor points based on at least the one or more moment-associating

elements. The one or more anchor points correspond to one or more timestamps respectively, and each anchor point of the one or more anchor points is navigable, searchable, or both navigable and searchable. In certain examples, the computer-implemented method further includes using the one or more anchor points to navigate the one or more pieces of moment-associating information based on at least the one or more timestamps. In some examples, the one or more anchor points include at least one selected from a group consisting of a word, a phrase, a photo, and a screenshot.

In some examples, the computer-implemented method further includes obtaining one or more moment-associating photos, the one or more moment-associating photos being one or more parts of the one or more moment-associating elements. The transforming the one or more moment-associating elements into one or more pieces of moment-associating information includes transforming the one or more moment-associating photos into one or more anchor photos, and the one or more anchor photos correspond to one or more timestamps respectively. In certain examples, each anchor photo of the one or more anchor photos is navigable, searchable, or both navigable and searchable. In some examples, the computer-implemented method further includes using the one or more anchor photos to navigate the one or more pieces of moment-associating information based on at least the one or more timestamps.

According to another embodiment, a system for receiving and processing one or more moment-associating elements includes a receiving module configured to receive the one or more moment-associating elements, a transforming module configured to transform the one or more moment-associating elements into one or more pieces of moment-associating information, and a transmitting module configured to transmit at least one piece of the one or more pieces of moment-associating information. The transforming module is further configured to segment the one or more moment-associating elements into a plurality of moment-associating segments, assign a segment speaker for each segment of the plurality of moment-associating segments, transcribe the plurality of moment-associating segments into a plurality of transcribed segments, and generate the one or more pieces of moment-associating information based on at least the plurality of transcribed segments and the segment speaker assigned for each segment of the plurality of moment-associating segments. For example, the system is implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15.

According to yet another embodiment, a non-transitory computer-readable medium with instructions stored thereon, that when executed by a processor, perform the processes comprising: receiving the one or more moment-associating elements; transforming the one or more moment-associating elements into one or more pieces of moment-associating information; and transmitting at least one piece of the one or more pieces of moment-associating information. The transforming the one or more moment-associating elements into one or more pieces of moment-associating information includes: segmenting the one or more moment-associating elements into a plurality of moment-associating segments; assigning a segment speaker for each segment of the plurality of moment-associating segments; transcribing the plurality of moment-associating segments into a plurality of transcribed segments; and generating the one or more pieces of moment-associating information based on at least the plurality of transcribed segments and the segment speaker assigned for each segment of the plurality of moment-

associating segments. For example, the non-transitory computer-readable medium is implemented according to at least FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, FIG. 9, FIG. 10, FIG. 11, FIG. 12, FIG. 13, FIG. 14, and/or FIG. 15.

For example, some or all components of various embodiments of the present invention each are, individually and/or in combination with at least another component, implemented using one or more software components, one or more hardware components, and/or one or more combinations of software and hardware components. In another example, some or all components of various embodiments of the present invention each are, individually and/or in combination with at least another component, implemented in one or more circuits, such as one or more analog circuits and/or one or more digital circuits. In yet another example, while the embodiments described above refer to particular features, the scope of the present invention also includes embodiments having different combinations of features and embodiments that do not include all of the described features. In yet another example, various embodiments and/or examples of the present invention can be combined.

Additionally, the methods and systems described herein may be implemented on many different types of processing devices by program code comprising program instructions that are executable by the device processing subsystem. The software program instructions may include source code, object code, machine code, or any other stored data that is operable to cause a processing system to perform the methods and operations described herein. Other implementations may also be used, however, such as firmware or even appropriately designed hardware configured to perform the methods and systems described herein.

The systems' and methods' data (e.g., associations, mappings, data input, data output, intermediate data results, final data results, etc.) may be stored and implemented in one or more different types of computer-implemented data stores, such as different types of storage devices and programming constructs (e.g., RAM, ROM, EEPROM, Flash memory, flat files, databases, programming data structures, programming variables, IF-THEN (or similar type) statement constructs, application programming interface, etc.). It is noted that data structures describe formats for use in organizing and storing data in databases, programs, memory, or other computer-readable media for use by a computer program.

The systems and methods may be provided on many different types of computer-readable media including computer storage mechanisms (e.g., CD-ROM, diskette, RAM, flash memory, computer's hard drive, DVD, etc.) that contain instructions (e.g., software) for use in execution by a processor to perform the methods' operations and implement the systems described herein. The computer components, software modules, functions, data stores and data structures described herein may be connected directly or indirectly to each other in order to allow the flow of data needed for their operations. It is also noted that a module or processor includes a unit of code that performs a software operation, and can be implemented for example as a subroutine unit of code, or as a software function unit of code, or as an object (as in an object-oriented paradigm), or as an applet, or in a computer script language, or as another type of computer code. The software components and/or functionality may be located on a single computer or distributed across multiple computers depending upon the situation at hand.

The computing system can include client devices and servers. A client device and server are generally remote from

29

each other and typically interact through a communication network. The relationship of client device and server arises by virtue of computer programs running on the respective computers and having a client device-server relationship to each other.

This specification contains many specifics for particular embodiments. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a combination can in some cases be removed from the combination, and a combination may, for example, be directed to a subcombination or variation of a subcombination.

Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

Although specific embodiments of the present invention have been described, it will be understood by those of skill in the art that there are other embodiments that are equivalent to the described embodiments. Accordingly, it is to be understood that the invention is not to be limited by the specific illustrated embodiments.

What is claimed is:

1. A computer-implemented method for processing a plurality of moment-associating elements, the method comprising:

segmenting a plurality of audio elements from the plurality of moment-associating elements into a plurality of audio segments;

transcribing, by an automated speech recognition (ASR) system using one or more acoustic models and one or more language models that are trained on data including one or more human speech audios, a first element of the plurality of audio segments into a first text element of a plurality of text elements, each acoustic model of the one or more acoustic model configured to recognize a specific type of sound;

transcribing, by the ASR system using the one or more language models, a second element of the plurality of audio segments into a second text element of the plurality of text elements;

identifying, by the ASR system using the one or more acoustic models and the one or more language models, one or more incorrectly transcribed words in the first transcribed text element;

updating the first transcribed text element by correcting the one or more incorrectly transcribed words in the first transcribed text element using one or more transcribed words in the second transcribed text element; and

30

generating one or more pieces of moment-associating information based on at least the plurality of audio segments and the plurality of text elements.

2. The computer-implemented method of claim 1, further comprising assigning a timestamp associated with each element of the plurality of moment-associating elements.

3. The computer-implemented method of claim 1, wherein the plurality of audio elements includes one or more voice elements of one or more voice-generating sources.

4. The computer-implemented method of claim 1, wherein the plurality of moment-associating elements further includes one or more visual elements or one or more environmental elements.

5. The computer-implemented method of claim 4, wherein the one or more visual elements includes at least one selected from a group consisting of pictures, images, screenshots, video frames, one or more projections, and holograms.

6. The computer-implemented method of claim 4, wherein the one or more environmental elements includes at least one selected from a group consisting of global positions, location types, and conditions associated with the one or more environmental element.

7. The computer-implemented method of claim 4, wherein the one or more environmental elements includes at least one selected from a group consisting of a longitude, a latitude, an altitude, a country, a city, a street, a location type, a temperature, a humidity, a movement, a velocity of a movement, a direction of a movement, an ambient noise level, and one or more echo properties.

8. The computer-implemented method of claim 1, further comprising:

receiving one or more voice elements of one or more voice-generating sources; and

receiving one or more voiceprints corresponding to the one or more voice-generating sources respectively.

9. The computer-implemented method of claim 8, further comprising:

segmenting the plurality of audio elements into a plurality of audio segments includes segmenting the plurality of audio elements into the plurality of audio segments based on at least the one or more voiceprints;

wherein the transcribing the plurality of audio segments into a plurality of text elements includes transcribing the plurality of audio segments into the plurality of text elements based on at least the one or more voiceprints.

10. The computer-implemented method of claim 8, wherein the receiving one or more voiceprints corresponding to the one or more voice-generating sources respectively includes receiving one or more acoustic models and/or language models corresponding to the one or more voice-generating sources respectively.

11. The computer-implemented method of claim 1, further comprising:

establishing one or more anchor points based on at least the plurality of moment-associating elements;

wherein:

the one or more anchor points correspond to one or more timestamps respectively; and

each anchor point of the one or more anchor points is navigable, searchable, or both navigable and searchable.

12. The computer-implemented method of claim 11, further comprising using the one or more anchor points to navigate the one or more pieces of moment-associating information based on at least the one or more timestamps.

31

13. The computer-implemented method of claim 11, wherein the one or more anchor points include at least one selected from a group consisting of a word, a phrase, a photo, and a screenshot.

14. The computer-implemented method of claim 1, further comprising:

obtaining one or more moment-associating photos, the one or more moment-associating photos being one or more parts of the plurality of moment-associating elements; and

transforming the one or more moment-associating photos into one or more anchor photos, wherein the one or more anchor photos correspond to one or more timestamps respectively.

15. The computer-implemented method of claim 14, wherein each anchor photo of the one or more anchor photos is navigable, searchable, or both navigable and searchable.

16. The computer-implemented method of claim 15, further comprising:

using the one or more anchor photos to navigate the one or more pieces of moment-associating information based on at least the one or more timestamps.

17. A system for processing a plurality of moment-associating elements, the system comprising:

a transforming module comprising at least one selected from a group consisting of one or more software modules or one or more processors, the transforming module configured to transform a plurality of moment-associating elements into one or more pieces of moment-associating information by:

segmenting a plurality of audio elements from the plurality of moment-associating elements into a plurality of audio segments;

transcribing, by an automated speech recognition (ASR) system using one or more acoustic models and one or more language models that are trained on data including one or more human speech audios, a first element of the plurality of audio segments into a first text element of a plurality of text elements;

transcribing, by the ASR system using the one or more language models, a second element of the plurality of audio segments into a second text element of the plurality of text elements;

identifying, by the ASR system using the one or more acoustic models and the one or more language mod-

32

els, one or more incorrectly transcribed words in the first transcribed text element;

updating the first transcribed text element by correcting the one or more incorrectly transcribed words in the first transcribed text element using one or more transcribed words in the second transcribed text element; and

generating one or more pieces of moment-associating information based on at least the plurality of audio segments and the plurality of text elements.

18. A computer-implemented method for receiving and processing a plurality of moment-associating elements, the method comprising:

receiving the plurality of moment-associating elements, the plurality of moment-associating elements including a plurality of audio elements;

establishing, by an automated speech recognition (ASR) system using one or more acoustic models and one or more language models that are trained on data including one or more human speech audios, one or more anchor points based on at least the plurality of moment-associating elements, the one or more anchor points corresponding to one or more timestamps respectively, and at least one of the one or more anchor points including an image that is searchable via a user-inputted image, each acoustic model of the one or more acoustic model configured to recognize a specific type of sound;

transforming, by the ASR system using the one or more language models, the plurality of moment-associating elements and the one or more anchor points into one or more pieces of moment-associating information; and

transmitting at least one piece of the one or more pieces of moment-associating information.

19. The method of claim 18, further comprising:

determining that the user-inputted image matches the image included in the at least one of the one or more anchor points by at least a predetermined threshold; and

navigating to at least one of the one or more pieces of moment-associating information based at least in part on the at least one of the one or more anchor points.

20. The method of claim 19, further comprising: transcribing the plurality of audio segments into a plurality of text elements, wherein the image that is searchable is presented in-line with the plurality of text elements.

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